ACCESS · INDIGENOUS SKILLS 4 SUCCESS

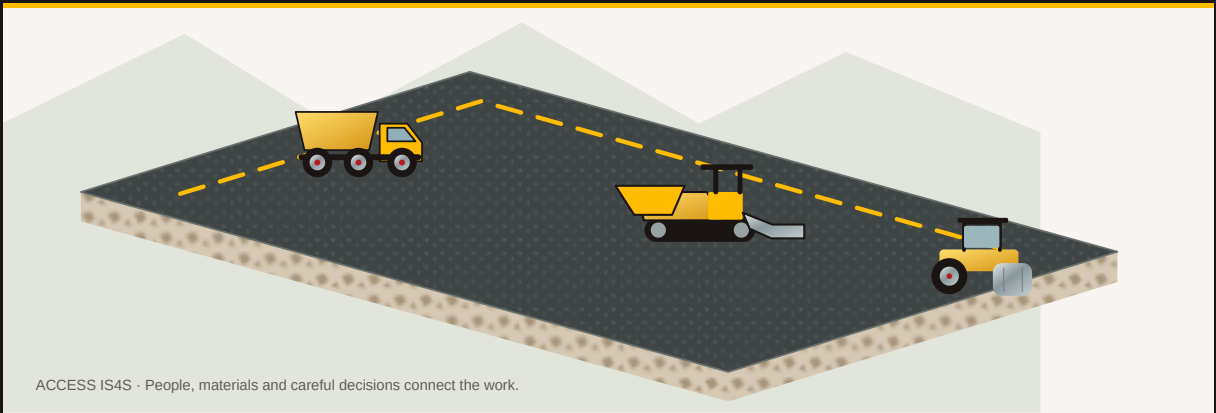
Asphalt paving · Foundations and safe work

Study textbook · Volume 1 · A1–A6 and B1

Classroom development edition · 30 October 2026 · Technical review and pilot pending

ACCESS IS4S / ASPHALT STUDY TEXTBOOK

Roadbuilding connections



ACCESS IS4S. (2026). Roadbuilding connections [Original cover illustration]. Conceptual scene.

Contents

1. Asphalt paving · Foundations and safe work
2. 1 · The industry and its work
3. 2 · How roads changed
4. 3 · From materials to paving
5. 4 · Working in a paving team
6. 5 · Read road documents and quantities
7. 6 · Investigate a paving trend
8. 7 · Recognize hazards and prepare for safe work
9. Working reference and evidence record

Asphalt paving · Foundations and safe work

A road is a connection between people

A delivery gets through. A family reaches a service. A worker returns home. Good paving depends on material knowledge, careful preparation and people who understand one another. This volume teaches you to read the work, name what you see, check a quantity and explain a safe next step.

The learning follows **A1–A6 and B1** of the SkilledTradesBC Asphalt Paving/Laydown Technician program outline. Seven chapters contain the readings, definitions, drawings, worked examples and practice information needed for this stage. Later volumes will develop B2–L6. This is the **30 October 2026 classroom development edition**; qualified technical review and classroom pilot are pending. Practical competence requires supervised instruction, observation and the applicable workplace authorization.

FIGURE T01

From rock to a road people use



Note. ACCESS IS4S. (2026). From rock to a road people use [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

Your route through the volume

Chapter	You will learn to...	Evidence to keep
1 · The industry and its work	Explain the material route, equipment, road purposes, industry numbers and project documents	A labelled process map and a checked industry brief
2 · How roads changed	Distinguish materials, methods, uses and historical events	A timeline and a careful public caption
3 · Materials, machines and placement	Trace mixture through a plant and paver; identify parts and functions	An annotated machine study and a material-flow explanation
4 · Working in a paving team	Connect roles, responsibilities, communication and learning	A handover and a development plan
5 · Read a project in three views	Read a plan, profile and section; check quantities and a revision	A calculation sheet and map-reading record
6 · Investigate a paving trend	Test a claim against an equivalent service and useful evidence	A comparison brief
7 · Recognize hazards and prepare for safe work	Connect PPE, exposure controls, orientation, WHMIS, first aid, rigging and public protection	A complete classroom safety brief

Use the nine Skills for Success every day

Skill	What it looks like in this book
Reading	Find a condition, symbol or component in the relevant document.
Writing	Leave a clear record that another person can use.
Numeracy	Keep units attached to quantities; estimate and check a result.
Digital	Open, inspect, save and reopen a file or map record on a device.
Problem Solving	Separate a known fact from a missing fact before deciding.
Communication	Explain, listen and confirm shared understanding.
Collaboration	Contribute to a shared task and check another person's work.
Adaptability	Revise a response when the conditions or evidence change.
Creativity and Innovation	Develop and test a clearer explanation or practical improvement.

Keep the activity identifier, your own work, feedback and your revision. A group answer needs a record of each learner's contribution. A paper plan can prepare for a digital task; the digital evidence comes from using the device. A written outcome needs text whose wording you control. Use accessible tools and instructor support as needed, and record that support.

Learn with care for people and place

Road work in B.C. takes place in lands and communities with different histories, rights, relationships and priorities. Ask whose access could change, whose knowledge is needed and who has authority to make a decision. A map cannot supply those relationships. Nobody in the class is expected to represent a Nation, disclose personal experience or share community-held knowledge. Contributions are voluntary; ask before recording or sharing them.

Our Cedar Learning Access case is fictional. Its community facility gives the work a clear purpose without assigning teachings, identities or permissions to a real Nation. Treat classmates' questions as useful information. A learner who identifies uncertainty and seeks the right support is contributing to the crew.

A practical study routine

Look → name → explain → check. Start with the illustration, name the trade feature, explain what it does, then check the caption and the reading. Try the activity before opening the answer guide. Keep an initial attempt and a revision so you can see what changed.

A figure may be an original teaching diagram, a redraw, a manufacturer crop or a screenshot. Its credit identifies the edition and page where relevant. Match a machine's model and serial range before using its operating instructions. General process pictures explain functions; actual equipment procedures and site conditions govern practical work.

Use the worked answers after your attempt. The figure register gives every image's credit and provenance. The interactive plan desk supports Chapter 5 and works offline with this repository.

1 · The industry and its work

SKILLEDTRADESBC · LINE A

A1 · Introduction to the paving industry and paving equipment

Read the trade words, use them in a clear explanation, then keep evidence of your own work.

Outline learning task	Practise here
A1.1–A1.5, A1.7 · Industry, equipment, resources, purposes and road types	TB1-01–04; Chapters 2–3
A1.6 · Demonstrate understanding of relevant statistics	LA1-01–02
A1.8 · Review relevant regulations and funding	LA1-03–04
A1.9 · Discuss environmentally friendly	LA1-05–06

Start with the purpose of the road

An asphalt paving/laydown technician helps turn a prepared site and a specified mixture into a pavement that serves its intended use. The work connects materials, machines, people and records. **Infrastructure** means the systems that support daily activity: roads are part of that network, alongside services such as drainage and utilities. A road can carry a delivery, support trade, provide access to school or health care, or serve recreation. The same route may serve several purposes.

The person using the finished road sees the surface. The crew also needs to understand what supports it. A suitable foundation, drainage, appropriate materials and verified construction all contribute to performance. A dark, smooth surface alone cannot tell you whether those conditions were met.

Cedar Learning Access, fictional case. A learning centre needs an access route that will serve people arriving on foot and in vehicles. The owner has appointed a designer and contractor. Before work starts, the crew needs current drawings, an agreed scope, a traffic and public-access arrangement, suitable equipment and a clear safety orientation. A neighbour's request about an entrance matters, but the crew must refer a proposed design change through the authorized project process. Listening and recording the concern helps the right person respond.

Follow the material and the information

Aggregate is mineral material such as sand, gravel or crushed stone. A **pit** is a source associated with excavating deposits such as sand and gravel; a **quarry** typically supplies rock that is extracted and processed. Crushing reduces particle size; screening separates material by size. Suitable material comes from an approved supply and must meet the project requirements. Finding nearby stone does not establish its suitability, extraction permission or environmental approval.

B.C. has varied landscapes and resource locations. Transport distance, the approved grading and properties of the aggregate, access and supply consistency affect a project. The crew uses the specified source and delivery records. A local material can be useful, but “local” is not a quality test. Equally, a long haul can change scheduling and logistics without changing the required product.

Asphalt binder, also called asphalt cement in the handbook, holds mineral particles together in an asphalt mixture. It is different from Portland cement, which is used in concrete. A plant proportions and processes the required materials. A haul vehicle brings the mixture to the paving operation. The paver receives and moves the mixture; its screed contributes to shaping the mat. Rollers compact the placed material, and the work is checked through the required inspection and testing process.

The information travels with the material. A delivery ticket identifies a load; a drawing identifies where the work belongs; a specification defines requirements; a test record reports an observation under a stated method. These records answer different questions. A ticket's mass cannot, by itself, prove the thickness or density achieved in the road.

TB1-01 · Build the material story

Draw five connected stages: source, production, delivery, placement and compaction/checking. At each stage, add one material or machine term, one purpose and one record or question. Explain the route to a partner using *aggregate*, *binder*, *paver*, *screed* and *compaction*. Your partner points to the stage being described, then you swap. Each learner keeps an independently labelled map. **Skills:** Reading, Communication, Collaboration, Creativity and Innovation.

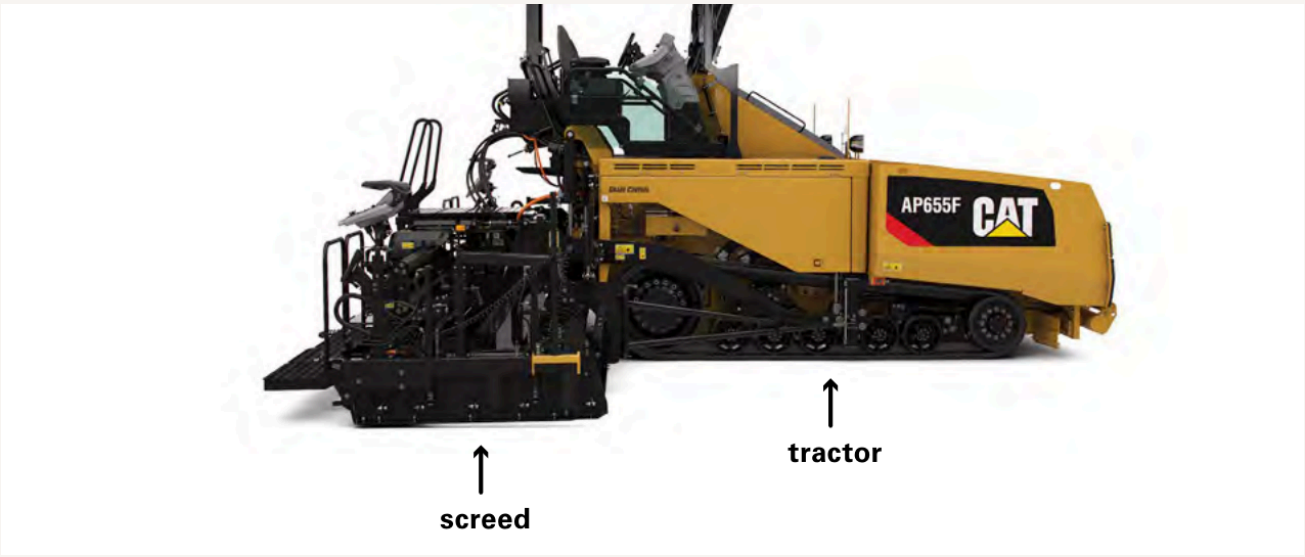
Name the equipment by what it does

Equipment or component	Main function in this introduction	A distinction to remember
Crusher and screen	Reduce and separate aggregate sizes in an appropriate processing operation	Processing equipment does not establish that the product meets a specification.
Asphalt plant	Produce the specified mixture through the applicable production arrangement	Batch and continuous processes differ; Chapter 3 follows both.
Haul truck	Transport material and support controlled delivery	A load ticket is a record, not a placement test.
Paver tractor	Provide propulsion, power and material-handling functions	The tractor and screed are the two main sections shown below.
Hopper and conveyors	Receive mixture and move it rearward through the paver	The conveying mechanism has moving parts and hazardous energy.
Augers	Distribute mixture across the working width ahead of the screed	Their motion differs from the rearward conveyor route.
Screed	Shape the mat as part of the paving system	It responds to several interacting factors; it is not a fixed-depth stamp.

Equipment or component	Main function in this introduction	A distinction to remember
Roller	Compact material through the applicable procedure	A smooth appearance or a pass count alone does not prove density.
Rake, lute and other hand tools	Support appropriate spreading, shaping and finishing tasks	Similar-looking tools have different heads, functions and limits.

FIGURE T02

Two main sections of the paver

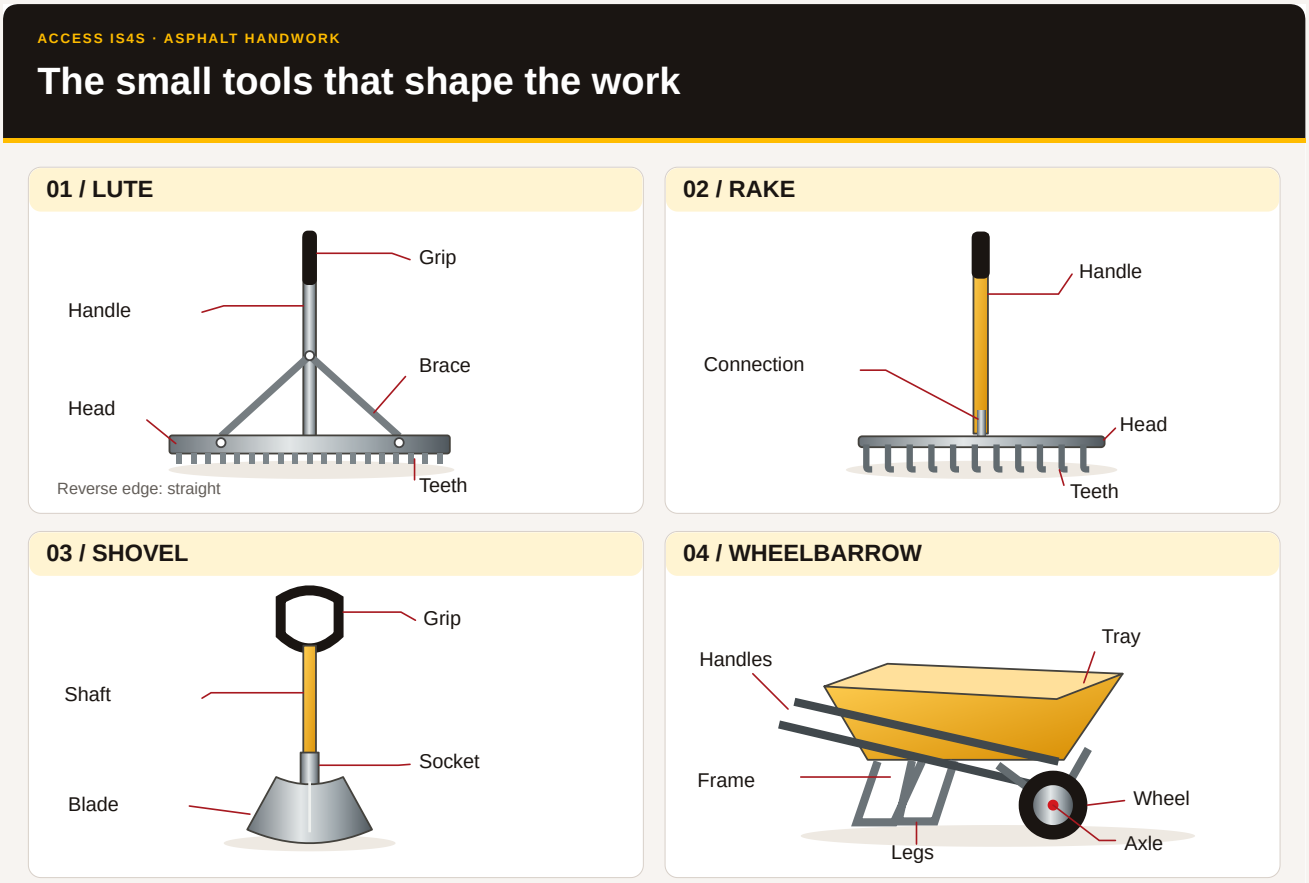


Note. From Guide to Asphalt Paving (QEBQ1888, source PDF p. 63), by Caterpillar, 2016. Copyright Caterpillar. Faithful 200 dpi colour crop; original diagrams, callouts and labels retained. No labels invented or replaced. Complete source-page link accompanies the figure. Scope: Conceptual paver example in the 2016 application guide; this picture does not identify a specific machine serial number. Source and context.

Look and explain. Locate the screed in the manufacturer picture. Then locate the tractor. Explain how a machine can have one section that moves and handles material and another that contributes to shaping the mat. These are functional categories; an actual machine may have additional systems and different controls.

FIGURE T03

Name the parts of the handwork kit



Note. ACCESS IS4S. (2026). The small tools that shape the work [Original instructional SVG]. Illustrative anatomy; confirm the actual model and parts list. Source and context.

A **lute** commonly has a broad head used in asphalt handwork. A **rake** has teeth suited to its intended material-handling task. Product terminology and head arrangements vary. Use the actual product description and instructions to distinguish a toothed head, a straight edge, a brace and a handle connection. A **shovel** has a blade and shaft, often with a grip; a **wheelbarrow** has a tray, handles, frame, wheel arrangement and supports. Loose connections, damaged handles, sharp damage or an unsuitable load can change the risk even when the tool looks familiar.

FIGURE T04

Match the drawing to real tools



Note. From Partner Resource Guide (source PDF p. 156; printed p. 155), by Seymour Midwest, 2026 publisher download. Copyright Seymour Midwest. Upper illustration area cropped; original product relationships retained. Source and context.

TB1-02 · Connect a part to a purpose

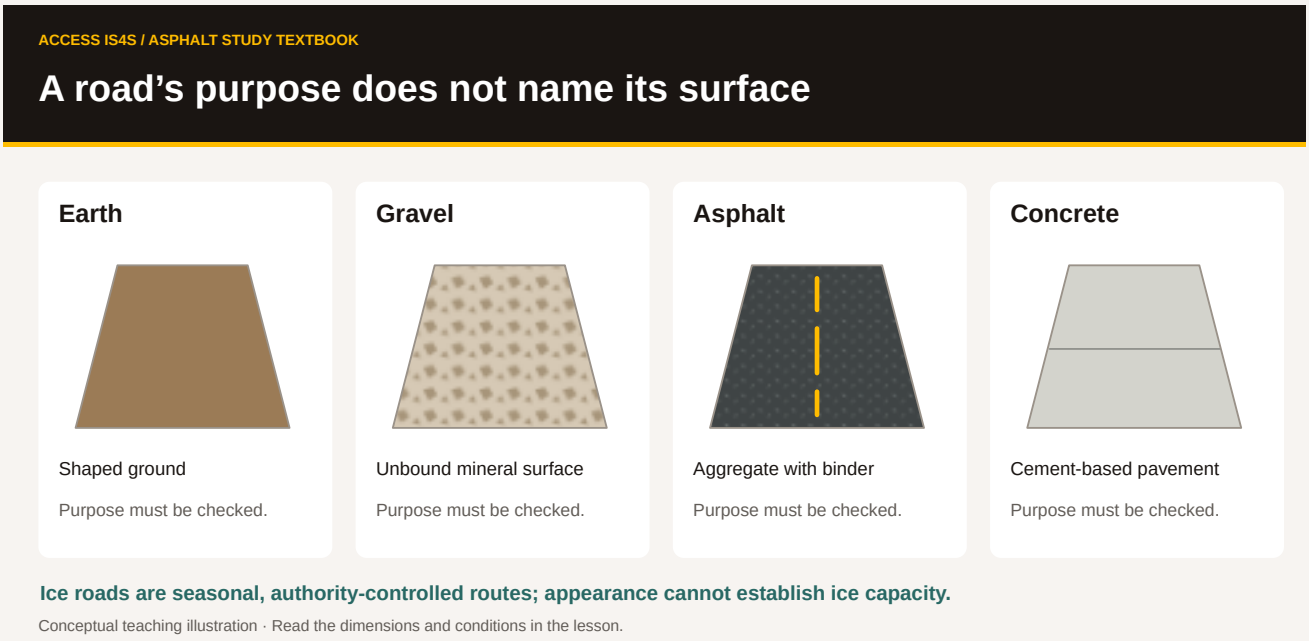
Identify tractor and screed in the CAT image. On the hand-tool diagram, identify a lute head, brace and handle; a rake's teeth; a shovel blade; and a wheelbarrow frame and wheel. Compare one diagram with the manufacturer's catalogue photograph. Record one feature that agrees and one feature that must be checked on the actual tool. Write a short defect report for the fictional observation “the rake head moves at its handle connection.” Describe what you observed and who should assess it before use; do not invent a repair. **Skills:** Reading, Writing, Problem Solving.

Roads have different surfaces and purposes

An **earth road** uses the ground surface, which may be shaped or improved. A **gravel road** has an unbound mineral surface. **Asphalt pavement** uses an asphalt-bound surface; **concrete pavement** uses a cement-based material. Supporting layers and drainage vary with the design. An **ice road** is a seasonal route whose opening, loading and use require the relevant authority's controls. A photograph does not establish safe ice capacity or permission to travel.

FIGURE T05

A road’s purpose does not name its surface



Note. ACCESS IS4S. (2026). A road’s purpose does not name its surface [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

A **resource road** describes an access context, often connected with resource activity; it does not name one surface. A **logging road** describes a use. A **corduroy road** describes a construction form using logs laid across the direction of travel. Those three terms are not interchangeable. Chapter 2 develops the history and the evidence for different methods.

TB1-03 · Classify carefully

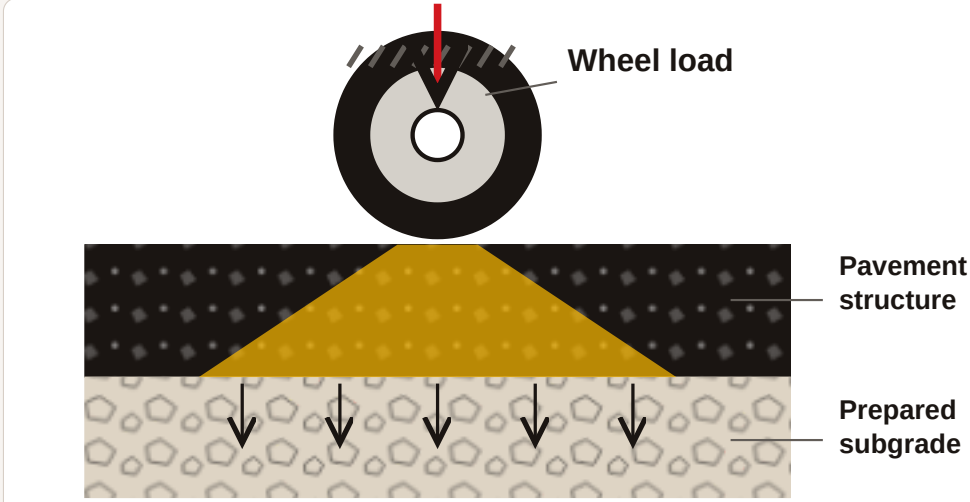
For each phrase, identify whether it primarily describes material/surface, construction method or use: gravel road; logging road; corduroy road; concrete pavement; recreational access; seasonal ice road. Some phrases carry more than one kind of information—explain your choice. Then write three ways a road can support a community and two questions about effects on people, land or water that would need project-specific discussion. Keep personal or community-held knowledge private unless its sharing is agreed. **Skills:** Reading, Writing, Problem Solving.

Explain why the foundation matters

A pavement carries loads through its structure to the prepared ground. The layers work as a system. Their materials, thicknesses and condition influence how the system responds. The illustration below shows a qualitative idea: the region carrying the load becomes wider through the pavement structure. It is not a pressure calculation or a design method.

FIGURE T06

Spread the load



Note. Adapted from Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4). Figure 1.5, p. 13 / PDF 47. Simplified qualitative section; replaced pressure profile with a shaded load-transfer area and plain-language labels. Copyright 2007 by Asphalt Institute. ACCESS redraw (2026). Source and context.

TB1-04 · Give a complete first explanation

Prepare a one-minute explanation for a new classmate: what asphalt paving is, why roads matter, how aggregate and binder reach the road, which equipment functions you can now identify, and why supporting layers need attention. Include one thing the pictures cannot establish about an actual job. Ask the listener to restate your explanation; revise a sentence that was unclear. Save and reopen a typed or recorded version on a device, with permission before recording another person. **Skills:** Communication, Adaptability, Digital.

Read a number with its label

A statement about an industry needs a measure, geography and period. A count of people employed is different from a count of employers. A forecast of openings over ten years is different from the number employed in one year. Construction includes building work and civil engineering as well as paving. A broad construction figure provides context; it cannot be renamed an asphalt-paving total.

Evidence card I1 · B.C. construction. WorkBC's Construction industry profile, retrieved 29 October 2026, reports the following. The 2024 values are attributed on the page to the Labour Force Survey. The openings forecast is attributed to the B.C. Labour Market Outlook.

Measure	Period	Reported value	Population or scope
Employment	2024	246,750	B.C. construction industry
Full-time share	2024	92%	People employed in B.C. construction
Part-time share	2024	8%	People employed in B.C. construction
Full-time share	2024	80%	People employed across B.C. industries
Part-time share	2024	20%	People employed across B.C. industries
Forecast job openings	2025–2035	87,280	B.C. construction industry

The public page does not supply an asphalt-specific employer count in this extract. Shares are published rounded values; multiplying them by an employment total produces an approximation, not a newly observed headcount. Forecasts describe expected demand under assumptions and do not guarantee a job or a person's hours.

Evidence card I1b · Road-industry business locations. Statistics Canada's table 33-10-1095-01, *Canadian Business Counts, with employees, December 2025*, released 13 February 2026, reports **427 active statistical locations with employees** in B.C. under **NAICS 237310, Highway, street and bridge construction**. This is a closer road-industry classification than all construction, but it includes more than asphalt paving. A company with several statistical locations can be counted several times. Locations are not a count of unique employers, people or projects.

Employment-size category	Reported B.C. statistical locations
1–4 employees	175
5–9 employees	63
10–19 employees	72
20–49 employees	62
50–99 employees	32
100–199 employees	18
200–499 employees	4

Employment-size category	Reported B.C. statistical locations
500 or more employees	1
Total with employees	427

The table's notes explicitly say not to use the size ranges to calculate total employment, and advise against using these counts as a time series because methodological changes can affect differences. The retained machine record has reference-period code `2025-07`; the publisher labels the semi-annual table **December 2025**. Keep both in a source record rather than silently rewriting the downloaded field.

Worked example. The difference between the construction full-time share and the all-industry share is `92% − 80% = 12 percentage points`. It is not a 12% increase in employment. An approximate construction part-time count would be `246,750 × 0.08 = 19,740 people`, using a rounded share. A careful sentence keeps the word *approximately* and the 2024 construction scope.

LA1-01 · Report what the data can support

Write a four-sentence industry note using I1: one employment statement; the full-time-share comparison; one forecast statement; and one limitation. Correct this claim: “There are 246,750 asphalt employers in B.C. and 87,280 guaranteed new paving jobs.” Then use I1b to check the sum of the eight size categories, find the difference between the 1–4 and 5–9 categories, and find the 1–4 category's share of all 427 locations to two decimal places. Explain why neither adding category midpoints nor subtracting 427 from 246,750 gives a valid workforce result. Mark the publisher, period, measure and scope beside each number. **Skills:** Reading, Numeracy, Writing.

LA1-02 · Retrieve and preserve a source record

Using the public WorkBC link in the research record, locate one I1 value on a device. Record the page title, publisher, address, access date, measure and reporting year. Compare it with I1; a changed page does not make the retained dated extract disappear. Save and reopen your note. If access is unavailable, use the printed card for reading and schedule a later digital turn. **Skills:** Digital, Reading, Adaptability.

Separate law, contract information and funding

Evidence card I2 · Workplace orientation. B.C.'s Occupational Health and Safety Regulation, sections 3.22–3.25, defines a new worker to include someone new to a workplace and people affected by changed workplace hazards. Section 3.23 requires the employer to provide workplace-specific health and safety orientation and training before a young or new worker begins work. Topics include supervisor contact information, rights and responsibilities, hazards, emergency procedures, and instruction and demonstration of the work task. Section 3.24 addresses additional training when observation shows it is needed or when requested; section 3.25 requires records. This card identifies selected provisions; it is not the complete orientation.

Evidence card I3 · Highway specifications. B.C.'s Standard Specifications for Highway Construction page, updated 1 April 2026, describes the specifications as a standard reference in construction contracts. It identifies the 2025 edition as adopted 1 February 2025 and an amendments appendix updated 1 April

2026. The page says modifications for local conditions are specified in special provisions for each contract. A current public edition is a starting point: the signed contract, drawings, amendments and special provisions establish which requirements apply to a particular job. A specification reference is not automatically a statute.

Evidence card I3b · A defined abbreviation. The 2006 *Design Build Standard Specifications for Highway Construction*, clause **502.03.01**, defines **EPS as End Product Specification**. In that clause, the contractor is responsible for construction-process quality control and the Ministry performs specified end-product audit sampling and testing for acceptance/rejection and payment adjustment. The definition is confirmed in PDF page 297, printed 1 of 22. That historical definition does not establish which edition governs a current contract, or prove the intended meaning of every unexplained use of EPS. In a geofoam discussion, EPS means expanded polystyrene; read the surrounding subject and definition.

Evidence card I4 · A completed project. The Kicking Horse Canyon Phase 4 project pages report completion on 31 July 2024. The project overview lists Government of Canada investment of **$215.19 million** and B.C. government investment of **$385.58 million**. The procurement page records the Province awarding the design-build contract to **Kicking Horse Canyon Constructors** on 6 November 2020. That team consisted of Aecon Group Inc., Parsons Inc. and Emil Anderson Construction. Funding is not the same as a contractor's profit or an individual worker's pay. The public summary is not the project's complete contract.

LA1-03 · Match a question to a document

For each question, choose I2, I3, I3b, I4 or “additional document needed,” and explain the limit: What training does a new worker need before starting? Which specification edition governs a particular signed contract? Who are the two reported public funders? What is the contractor's profit? What does EPS mean in clause 502.03.01 of the 2006 specifications? What further context is needed to interpret EPS in an old note that gives no definition? Create a document register with title, issuer, date/version, status and question still open. **Skills:** Reading, Problem Solving, Writing.

LA1-04 · Check the funding statement

Add the two I4 contributions and calculate the federal share of that sum to two decimal places. Explain why “more than $600 million” is consistent with the listed values. Rehearse a 45-second explanation distinguishing funder, contracting authority and contractor. Your partner checks the roles and units, then you swap. **Skills:** Numeracy, Communication, Collaboration.

Compare environmental claims within a boundary

Asphalt and concrete are pavement materials, not complete environmental outcomes. Compare a defined service, time period and system boundary. A material-production figure alone leaves transport, construction, maintenance, use and end-of-life outside the answer. A reusable material is not automatically the option with the lowest total impact. The amount recovered, replacement of new material, processing energy, transport and required performance all matter.

Evidence card I5 · Material recovery, a bounded comparison. The U.S. EPA's March 2004 *Hot Mix Asphalt Plants* chapter describes asphalt mixtures that may include reclaimed asphalt pavement, or RAP, alongside aggregate and asphalt cement. FHWA's undated historical *Recycled Concrete Aggregate* review describes concrete being recycled as aggregate, principally for base material in the review's U.S. context. These are two evidenced recovery pathways. Neither card provides a matched B.C. life-cycle emissions comparison, equal pavement designs, haul distances or a common service life. They support "both materials can enter recovery pathways," not "either material is always greener." The references describe mechanisms rather than current local market shares.

Practice decision brief, fictional. A community facility is considering two pavement options for the same 50 m by 6 m access route over a proposed 20-year comparison period. Both must meet the same approved function and loading requirements, which a qualified designer would establish. Option A proposes asphalt with some RAP; Option B proposes concrete with some recycled aggregate. The proposer supplies no verified quantities, maintenance plan, emissions factors or transport distances. The group must decide whether a claim of "30% lower whole-life emissions" has been established. No numerical environmental advantage is supplied as fact.

LA1-05 · Make the strongest supported comparison

Create a comparison grid with material production, transport, construction, maintenance/use and end-of-life rows. For each option, distinguish the recovery pathway supported by I5 from information missing in the fictional brief. Write a conclusion of 60–90 words. Include the common function, 20-year proposed period, one supported similarity and at least three data needs. **Skills:** Reading, Problem Solving, Writing.

LA1-06 · Revise a claim when the scope changes

A proposer now says the "30%" applies only to factory production per tonne and compares different product quantities. Explain why that does not answer the whole-route, 20-year question. In pairs, redesign the comparison request so the next submission could be evaluated. Each learner records one contribution and one revised sentence. **Skills:** Numeracy, Collaboration, Adaptability, Creativity and Innovation.

Keep your industry note, source record, document register, funding calculation and comparison request. An instructor checks the evidence and the questions you leave open, not whether you force a confident answer from incomplete information.

2 · How roads changed

SKILLEDTRADESBC · LINE A

A2 · Describe history of asphalt pavement

Read the trade words, use them in a clear explanation, then keep evidence of your own work.

Outline learning task	Practise here
A2.1 · Describe evolution of roads	LA2-01, LA2-06
A2.2 · Describe products used	LA2-02, LA2-05
A2.3 · Describe logging roads	LA2-02–03, LA2-05
A2.4 · Explain methods of placement	LA2-05
A2.5 · Rationale for more roads	LA2-04, LA2-06

Ask what a road was built to do

Road history is a history of materials, movement, power and relationships with place. A route can serve travel, trade, access to resources, settlement or military movement. Its purpose can change. A road that enables one activity may also affect land, water, access and existing travel routes. A timeline should identify whose account is being used and avoid treating one engineering tradition as the beginning of all movement.

In B.C., a classroom discussion of road history does not give permission to map or publish community-held routes or knowledge. Use the public evidence cards below. Invite local knowledge only through the appropriate relationships and agreed purpose. The lesson asks you to explain change, not to speak for a community.

Say what kind of word it is

Trade word	Everyday explanation	Avoid this mix-up
Corduroy road	A construction form using closely laid logs across the direction of travel	The pattern describes a method; it does not prove a logging purpose
Logging road	A road used in forestry access or timber movement	It can have different surfaces and construction methods
Placement	Putting the prepared material into its intended position and form	Delivery gets it to the work; placement puts it in the layer

Trade word	Everyday explanation	Avoid this mix-up
Compaction	Making the placed material denser through the applicable process	A smooth-looking surface alone does not prove specified density

Try the words aloud. “Timber is the material; corduroy is the construction form; logging is one possible purpose.” Use this three-part pattern in LA2-02.

Evidence cards for a timeline

H1 · Via Appia. UNESCO describes the Appian Way as a major Roman road constructed and developed from **312 BCE to the fourth century CE**. Its initial strategic purpose included military conquest. It later supported connected settlements, agriculture and trade. Engineering works included the road and associated structures. One road can have several purposes over time; “Roman” does not mean “the world's first road.”

H2 · Alaska Highway. Public Services and Procurement Canada's history, updated **16 December 2025**, explains that the U.S. Army Corps of Engineers built the road during the Second World War to provide land access to Alaska. It identifies local First Nations' vital role as guides. The pioneer road was completed in **1942**, and the permanent location in **1943**. Responsibility for the Canadian portion transferred to National Defence in **1946**. Official civilian non-military use began in **1948**. Further responsibility changes occurred in **1961**, **1964** and **1992**. These are different events, not competing dates for the same event.

H3 · Post-war highways. FHWA identifies the U.S. Federal-Aid Highway Act of **1956** as a major step in the Interstate system. Its historical discussion connects the system with national planning and everyday movement of goods and people. The 1950s were within the Cold War period. A date and a national-security context do not prove that every new road had one motive; use evidence for the particular project.

H4 · Present resource roads. B.C.'s resource-road information describes a network used for resource industries and access. A resource road is not necessarily paved, and a logging road is not necessarily a corduroy road. Its owner, administrator, condition and access rules need to be established from the relevant record. A present-day category does not describe every historical method used to build it.

FIGURE T07

Dates tell different stories



Note. ACCESS IS4S. (2026). Dates tell different stories [Original instructional SVG]. Based on the public historical accounts by Public Services and Procurement Canada (2025), UNESCO (2024) and FHWA listed in the references. Separate time scales; no universal sequence implied.

LA2-01 · Build a timeline that preserves meaning

Place 312 BCE, 1942, 1943, 1946, 1948 and 1956 in chronological order. Label each event and its evidence card. Identify two different meanings of “completed” in H2. Write one sentence explaining why a timeline alone cannot show every social or environmental effect of roadbuilding. **Skills:** Reading, Writing, Problem Solving.

Materials and methods solve different problems

An earth route uses the ground surface, sometimes shaped or improved. Stone and gravel add mineral material. Cut or shaped stone can form a durable surface where the design, foundation and maintenance suit the use. Wood has also been used: a **corduroy road** is a historical construction form using closely laid logs across the travel direction, often associated with soft or wet ground. The log pattern explains the name. It is a method, while *logging road* describes a road's use; the categories overlap in some histories but are not synonyms.

Asphalt pavement combines aggregate with a bituminous binder. Concrete pavement uses a cement-based binding system. These materials have different production and construction processes. Do not confuse asphalt cement—the binder in an asphalt mixture—with Portland cement used in concrete. A road may have several layers and may be repaired with a different material from its original surface. Appearance alone is not enough to identify its entire construction.

Modern asphalt placement commonly involves delivery of mixture, controlled placement through a paver and screed, and compaction using rollers, with associated preparation, checks and finishing. Mechanization changes how material is moved and shaped; it does not remove the need for a suitable foundation, drainage, specifications or skilled people. This unit describes the change in method. Operating settings belong to the relevant supervised training and approved documents.

Worked example · Correct a category mistake. A museum draft says, “All logging roads were corduroy roads, and modern roads replaced wood with asphalt.” The first clause confuses purpose with method. The second implies one universal sequence. A better label is: “Corduroy construction used logs across a route. It was one historical way of addressing difficult ground. Logging roads have used varied surfaces and construction methods; modern asphalt paving is one distinct technology.”

LA2-02 · Match material, method and purpose

Create three columns headed material, method and purpose. Sort these terms: earth; cut stone; gravel; timber logs; concrete; moving harvested timber; military access; asphalt mixture; closely laid crosswise logs; delivery/placement/compaction; travel to a community facility. Explain the difference between asphalt cement and Portland cement. Explain why a purpose can be served by more than one material. Add one question you would ask before classifying an unfamiliar road from a photograph. **Skills:** Reading, Communication.

LA2-03 · Explain a road without inventing its history

Fictional archive label: “Photo R17, date not recorded. A line of logs crosses a wet-looking route. A truck appears in the background. No owner, purpose or place is recorded.” Write a 50–80-word description separating observation, reasonable interpretation and unknown information. A partner reads it and identifies any unsupported claim. Revise one sentence. **Skills:** Writing, Collaboration, Adaptability.

Explain why demand changed

War can accelerate road construction, as H2 shows. Post-war planning and Cold War concerns formed part of the setting for major highway development, while trade, industry, settlement and daily access also influenced demand. The same road can support freight, recreation and local travel. “More roads” is not a complete measure of benefit: maintenance, alternatives, land use and the people affected belong in the question.

LA2-04 · Compare two explanations

Compare these claims: “The Alaska Highway began as a wartime access project” and “All highways were built only for war.” Use H2 and H3 to support, limit or reject each claim. Calculate the interval from the 1942 pioneer-road completion to the 1948 opening to civilian non-military use. Explain why that interval does not mean no work occurred between those dates. **Skills:** Reading, Numeracy, Problem Solving.

LA2-05 · Connect a historical method to a modern sequence

Make a two-panel classroom display: one panel explains a corduroy route using the supplied definition; the other explains modern asphalt delivery, placement and compaction. Label material, method, purpose and one unknown in each. Use words, a simple sketch or an accessible digital layout. Do not provide a construction recipe. Each partner explains one panel and checks the other's labels. **Skills:** Creativity and Innovation, Collaboration, Communication; Digital if a tool is actually used.

LA2-06 · Independent transfer: a careful public caption

Write a 90–120-word caption titled “Roads change as needs and methods change.” Include a dated example, a material/method distinction, a reason for road growth and a boundary on what the evidence establishes. Add a source note from the evidence register. Keep the first and revised versions. Your next step may be a local public-archive enquiry, but record it as a proposed enquiry until evidence is actually obtained.

3 · From materials to paving

SKILLEDTRADESBC · LINE A

A3 · Describe paving technology

Read the trade words, use them in a clear explanation, then keep evidence of your own work.

Outline learning task	Practise here
A3.1 · Describe two types of asphalt plants	LA3-02
A3.2 · Describe asphalt ingredients	LA3-01, LA3-05
A3.3 · Discuss required resources	LA3-01
A3.4 · Describe liquids used in asphalt production	LA3-03
A3.5 · Describe asphalt recycling technology	LA3-04, LA3-06
A3.6 · Describe the asphalt “Train”	LA3-04, LA3-06

Follow the material through the system

Paving technology connects material sources, manufacturing, transport and placement. Begin with the function of each stage. A pit or quarry supplies mineral material; processing prepares aggregate to specified characteristics; a plant combines selected materials; haul vehicles move the mixture; paving equipment places and shapes it; rollers compact the placed layer. Preparation, testing, traffic arrangements and finishing also matter. A flow map is a way to understand relationships, not permission to operate equipment.

Figure LA-1

Follow the material, step by step



Note. An original ACCESS IS4S instructional illustration. The process distinctions are explained in this chapter; the drawing supplies no operating settings or mix proportions. By ACCESS IS4S (2026a); see the figure references in the working reference section.

Read the trade word, then explain its job

Trade word	Plain-language meaning	Say it in context
Asphalt cement / asphalt binder	The binding material in an asphalt mixture	“The binder holds the aggregate together.”
Processed gravel / aggregate	Mineral material prepared for the required characteristics	“The aggregate source still needs to meet the specification.”
Batch plant	Makes identifiable batches of mixture	“This record identifies one batch.”
Drum-mix plant	Uses a continuous material and mixing process; layouts vary	“This record describes a continuous feed period.”

Trade word	Plain-language meaning	Say it in context
Pit / quarry	Sources of mineral material: unconsolidated deposits / rock	“The source name does not prove material suitability.”
Asphalt train	A coordinated set of equipment and roles	“Name the process before explaining the train.”

The outline uses **asphalt cement**. This book also uses **asphalt binder** for that material. Keep both terms available when reading different documents; neither means Portland cement. Use the complete phrase when the short word *cement* could be misunderstood.

Know what the ingredients do

Aggregate is mineral material such as processed sand, gravel or crushed stone. A **pit** generally obtains unconsolidated deposits; a **quarry** extracts rock that may need crushing and screening. Material availability does not establish suitability: the required properties come from the approved design and specification. A short haul distance alone cannot answer whether a source meets the project requirements.

Asphalt binder, also called asphalt cement in many documents, binds the aggregate in asphalt mixtures. It is distinct from the Portland cement used in concrete. **Additives** and modified binders can change mixture characteristics for a defined purpose, but their selection and proportions require the applicable approved information. Fuel used to heat a production process is not automatically an ingredient in the pavement mixture. A delivery label must identify the actual product rather than relying on the loose word *oil*.

Technology evidence card T1. EPA's *Hot Mix Asphalt Plants* description identifies sized aggregate, liquid asphalt cement and potentially reclaimed asphalt pavement as components of hot-mix asphalt. It describes batch and drum-mix production. Its technical account is used here for the process distinction; its historical U.S. production totals and operating details are not current B.C. specifications.

LA3-01 · Explain the ingredients to a new learner

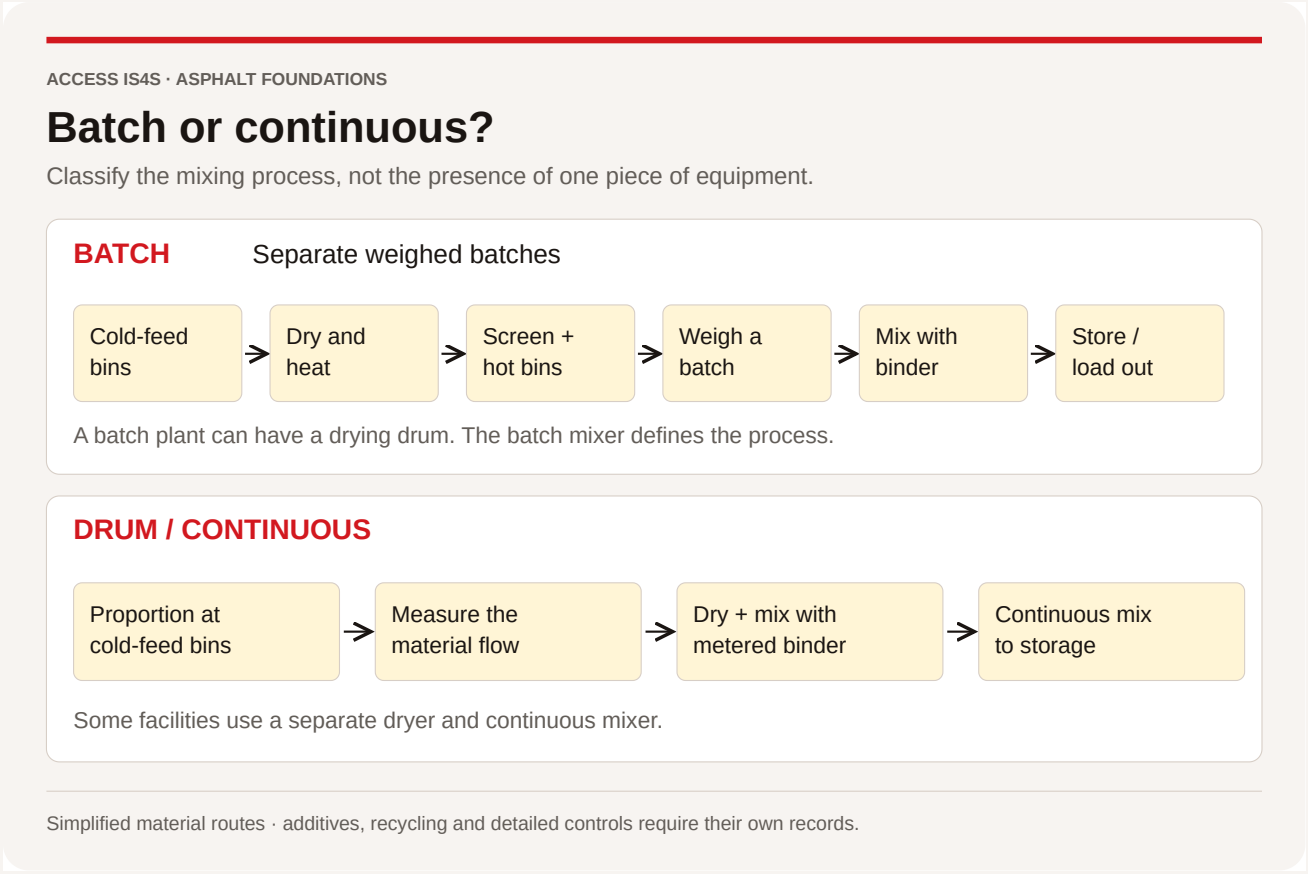
Write a three-sentence explanation that distinguishes aggregate, asphalt binder and fuel. Include the outline term *asphalt cement* and explain its relationship to *asphalt binder*. Add a sentence explaining the conditional purpose of an additive and why its use needs approved information. Distinguish a pit from a quarry, then name one question about an aggregate source that proximity alone cannot answer. Explain why “cement” needs context when comparing asphalt and concrete. **Skills:** Reading, Writing, Communication.

Distinguish batch and drum production

A **batch plant** prepares identifiable batches: aggregate is proportioned and combined with binder in a mixer. A **drum-mix plant** uses a continuous process in which material flows through the plant. Drum designs differ; do not treat every drum layout as identical. The beginner distinction is batch quantities versus a continuing material stream, not “one uses a drum and the other has no drum.” A batch plant can have a drying drum before its mixer.

Figure LA-4

Batch or continuous?



Note. Adapted from *The asphalt handbook* (7th ed., MS-4; Figure 7.2, p. 310 / PDF 344; Figure 7.4, p. 312 / PDF 346), by Asphalt Institute, 2007. Converted equipment drawings into two simplified process lanes; omitted settings and detailed control systems. Original Figure 7.2 is credited to Terex Roadbuilding in the handbook's contributor list. Original Figure 7.4 is credited to National Highway Institute (NHI) in the handbook's contributor list. Copyright 2007 by Asphalt Institute; original contributor credits retained.

Fictional classroom production records. Plant B's teaching log lists batch 01, batch 02 and batch 03, each with a separate record of ingredients and release. Plant D's teaching log records a continuous feed period and time-stamped checks. These are deliberately simplified descriptions. Neither log provides an approved mix design, production rate or operating instruction.

Worked example. “Plant B has a drying drum, so it must be a drum-mix plant” is an invalid inference. Ask where and how the mixture is produced. B's separate batch records support a batch process in this case. The physical presence of one part does not classify the whole system.

LA3-02 · Compare two process records

Identify the production type indicated by each fictional log. Draw or describe the common stages and the key difference. Add a correction to the worked-example claim in your own words. In pairs, one learner explains the map while the other checks whether it accidentally implies that a batch plant lacks a drying drum. Swap roles. **Skills:** Reading, Problem Solving, Collaboration.

Treat product codes as information to verify

A product code can identify a grade system or a product, but it does not supply the whole application instruction. A penetration-grade designation and a performance-grade designation are different ways of describing binder characteristics. The numbers are not interchangeable merely because both appear beside asphalt. The purchaser and operator need the approved product, governing specification, supplier record and relevant safety information.

Fictional document-check case. An archive card contains the labels “Group A / C,” “85–100 Pen” and “PG 72–22,” and a handwritten note says “altered oil.” A new practice request says only “send the usual oil.” The meaning of “Group A / C” is not defined in this archive card. No current approved binder, project specification or supplier product sheet has been attached. These labels are for recognition and clarification; the case does not recommend either grade for B.C. work. Do not infer a mixing temperature, quantity or handling procedure from them.

LA3-03 · Write the clarification before the order

First identify which archive wording is undefined and must be clarified; do not guess a classification from the letters. Make a four-item information request that would identify the intended product and its governing documents. Explain why “usual oil” and an archive grade label are insufficient. Read your request to a partner, who checks whether you have accidentally approved a product. Keep it as a classroom draft. **Skills:** Writing, Communication, Problem Solving.

Separate the recycling routes

Reclaimed asphalt pavement (RAP) is asphalt pavement material recovered for possible reuse. A plant-recycling route takes recovered material through suitable processing and an approved production process. Reuse is conditional on the design and quality requirements. A claim that all recovered material can be added without testing is unsupported.

Hot-in-place recycling treats an existing asphalt surface on the road using a coordinated equipment train. Its broad purpose is to reuse existing material in place through a specialist process. It is different from hauling recovered material to a plant. A milling machine's removal of material is also not proof that the removed material has been successfully recycled. Track the actual route and evidence for reuse.

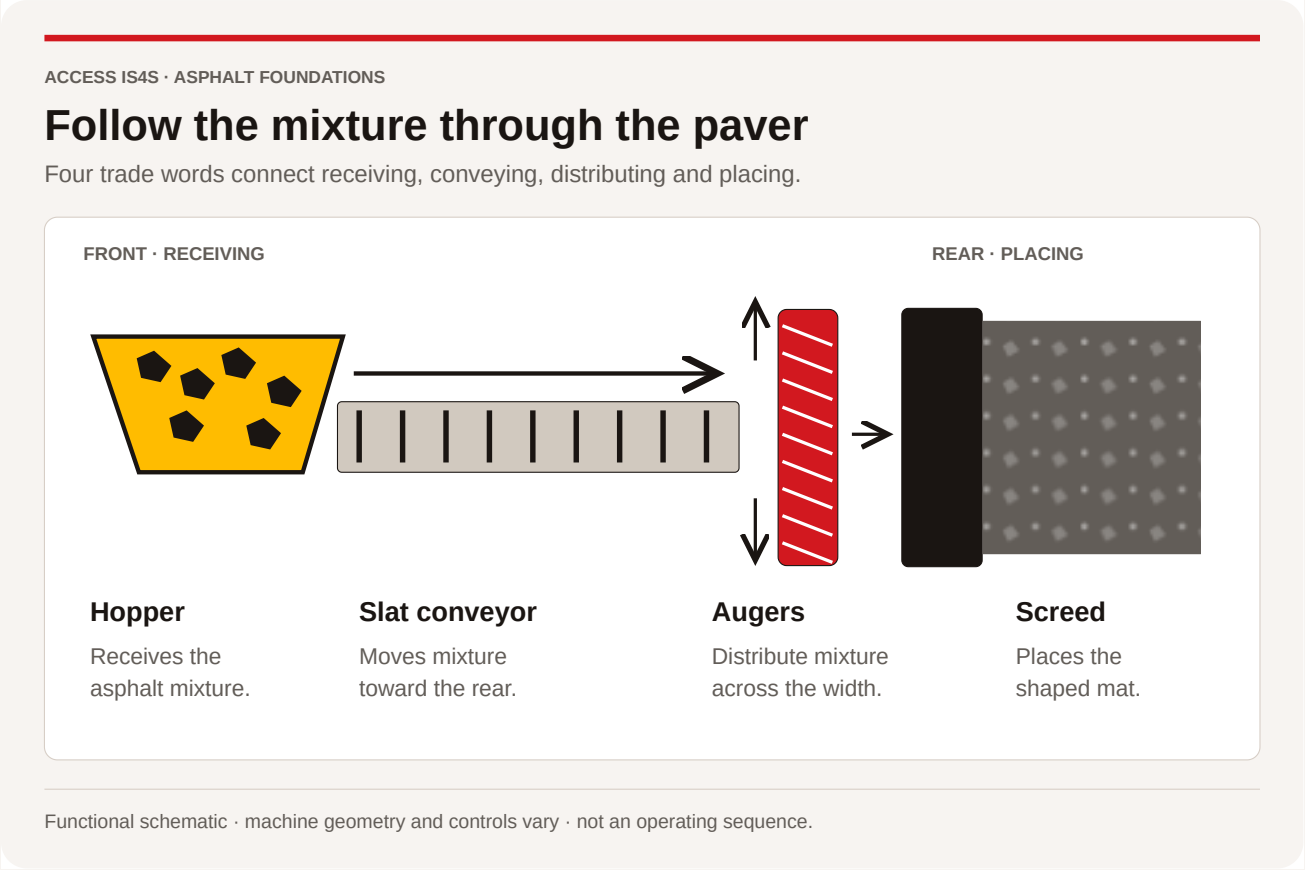
An **asphalt train** is a coordinated sequence of equipment and roles. In ordinary placement, haul, paver/screed and compaction functions connect. A specialist hot-in-place train has its own process and equipment. The word *train* does not mean every paving job uses the same machinery, controls or sequence.

Follow the mixture inside the paver

The **hopper** receives mixture. **Slat conveyors** move it toward the rear. **Augers** distribute it across the width ahead of the **screed**, which places and shapes the mat. The screed also performs the placement stage's initial compaction; rollers have a later compaction role. Read each noun together with its action. The material route explains a general function; an identified machine manual supplies the actual layout and controls.

Figure LA-5

Follow the mixture through the paver



Note. Adapted from *The asphalt handbook* (7th ed., MS-4; Figure 8.11, p. 377 / PDF 411; 8.4.1, component descriptions, p. 378 / PDF 412), by Asphalt Institute, 2007. Replaced the perspective equipment schematic with a labelled functional material route. Not drawn to scale or tied to an actual model. Original Figure 8.11 is credited to Volvo Construction Equipment in the handbook's contributor list. Copyright 2007 by Asphalt Institute; original contributor credits retained.

Worked explanation. "The conveyor moves mixture rearward; the auger distributes it across the width." The direction and job differ. Find the hopper and screed in the manufacturer photograph next, then explain why a general route diagram cannot establish another machine's numbered callouts.

Read one machine picture carefully

The flow map names a function. A manufacturer picture helps you locate a particular machine's parts. Start with the model and numbered callout, then read the matching name. This example introduces paver component vocabulary for later equipment study.

Figure LA-3

Find the named part on a LeeBoy 8680 paver

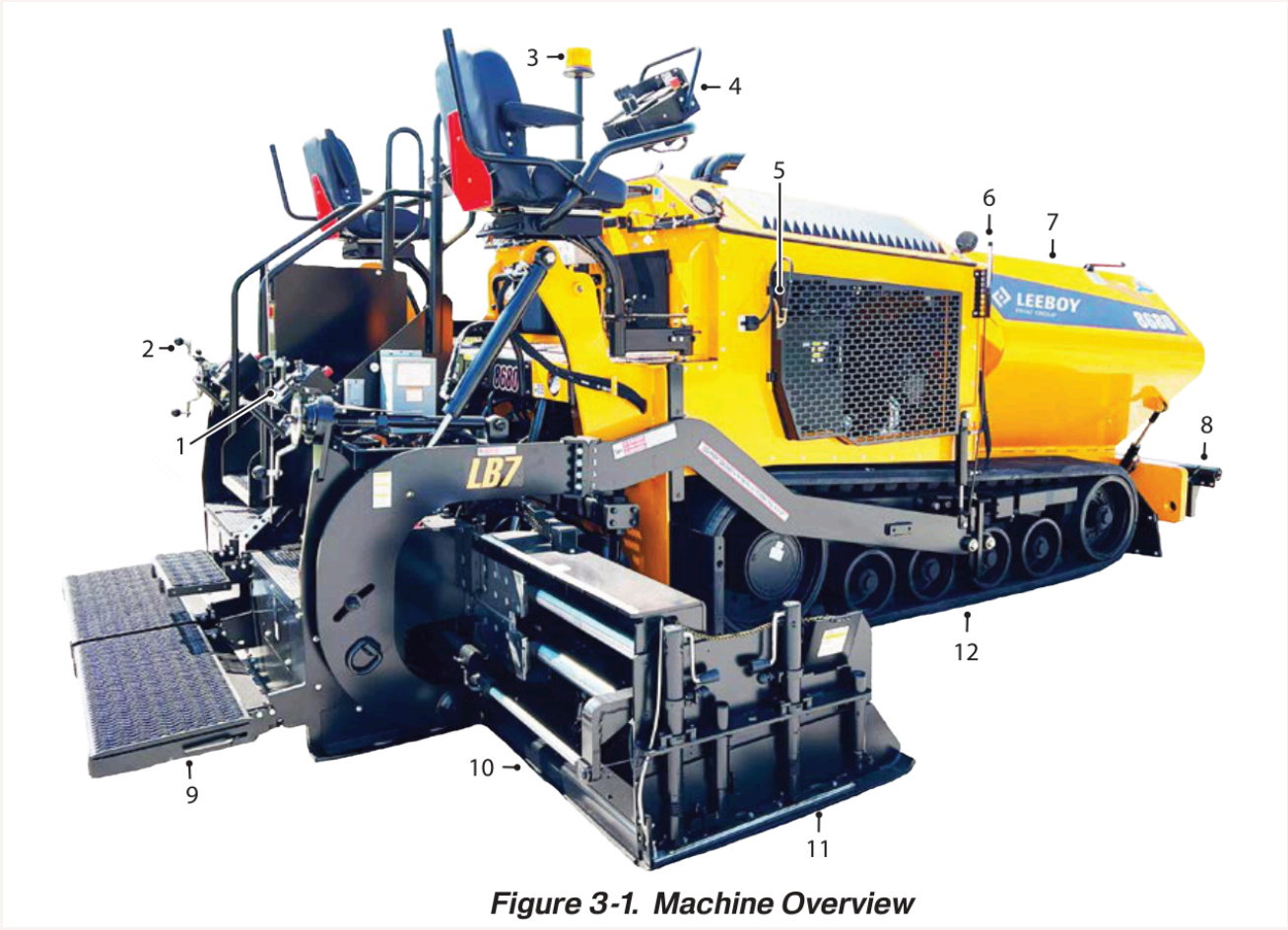


Figure 3-1. Machine Overview

Model 8680 · Manual 1033428-SOP · Serial number 365152 and above

Note. Cropped from Figure 3-1, "Machine Overview," in *LeeBoy model 8680 commercial paver: Safety and operations manual* (Manual No. 1033428-SOP, p. 3-7; PDF p. 43), by LeeBoy, 2026 (manufacturer publication). Copyright 2026 by LeeBoy. Original numbered callouts retained; the surrounding page and legend are omitted from the crop. Selected original component names appear below; the plain-language explanations are authored for this lesson. Open the complete source page and legend.

Original callout	Manufacturer's name	Plain-language explanation
7	Hopper	Holds the supplied asphalt material on this paver
10	Screed Extension	Extends the screed's working width
11	End Gate	Contributes to forming the edge of the placed mat
12	High-Speed Rubber Tracks	The rubber tracks on which this machine travels; the source name supplies no permission to select a travel speed

Worked example · Look → name → explain. "Callout 7 points to the **hopper**. It holds the supplied asphalt material. The photo does not tell me the approved mixture, the operating setting or whether another paver has the same arrangement." Use the selected table to explain a second part. Keep the manual identity with your answer. The complete manual has the other callouts and their context.

LA3-04 · Trace two reuse pathways

Draw two labelled routes: recovered material sent to a plant, and a specialist hot-in-place process on the existing road. Mark where the existing material is recovered or reused. Add one document or check needed before calling either route approved. Explain why “milled” and “recycled into a new pavement” are different claims. **Skills:** Reading, Creativity and Innovation, Problem Solving.

LA3-05 · Calculate a fictional material record

A classroom demonstration model uses **80 counters for new aggregate, 15 for recovered mixture and 5 for binder**, for 100 counters in total. Counters are a visual model, not a mix recipe; the recovered mixture already contains its own materials. Find each category's share of the model and check the sum. Explain why adding the five binder counters to the 15 recovered-mixture counters does **not** establish the real binder content. **Skills:** Numeracy, Reading.

LA3-06 · Respond to a changed process description

A team initially labels a photo “RAP going to a plant.” The caption is corrected to “specialist hot-in-place train.” Revise the material-flow card and give a handover identifying what changed, what remains accurate and what further process information would be needed. Each learner makes one recorded revision. **Skills:** Adaptability, Collaboration, Communication.

LA3-07 · Make a retrievable technology note

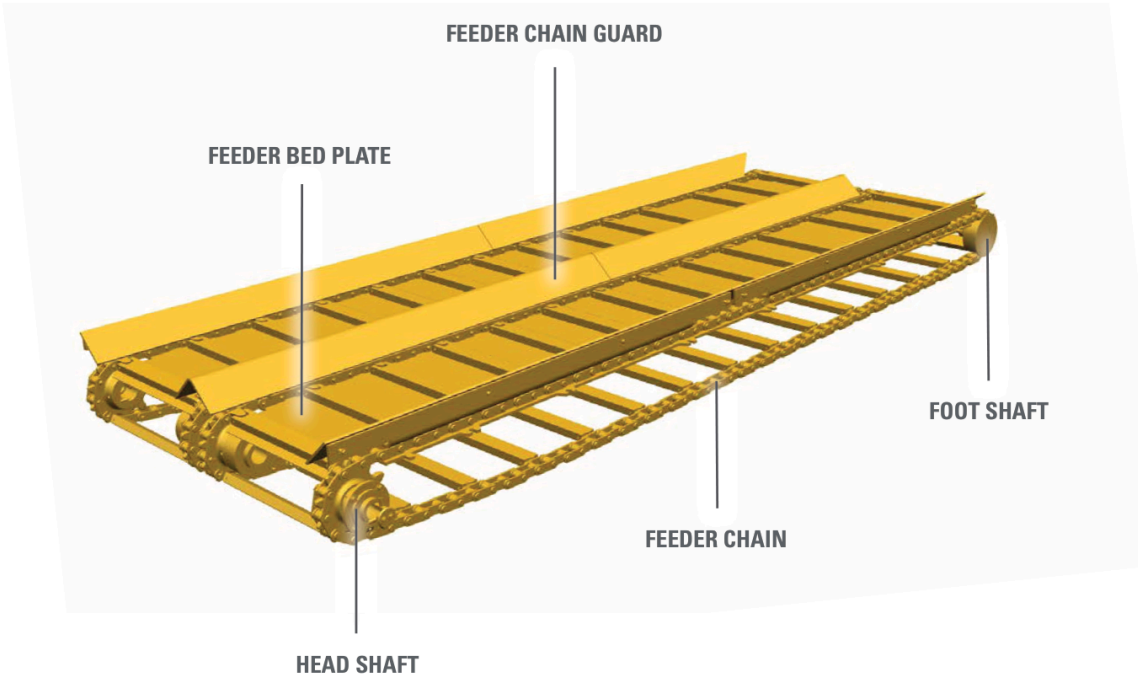
Create a one-page document with the two plant distinctions, ingredient roles, two recycling routes and the unresolved product request. Save, close and reopen it; check that another learner can find its title and version. Offer a paper route for composing, while keeping actual Digital evidence separate. Finish by explaining one function without providing an operating setting. Add a small Figure LA-3 note: choose two of callouts 7, 10, 11 or 12, record the manufacturer’s names, explain each in plain language, and retain the model, manual number and page. State one thing the photograph cannot establish. Add the four-stage material route from Figure LA-5, using the words hopper, slat conveyors, augers and screed. Explain how the conveyor and auger functions differ. Use Figure LA-4 to check that your plant comparison describes the mixing process.

Inspect the route at component level

The **feeder system** moves mixture through the paver. A feeder chain, its shafts and the bed plate have different functions: a chain transmits movement, shafts support or drive the assembly as designed, and the bed plate forms a wear surface. The **auger assembly** moves mixture across the width near the screed. The terms below belong to the manufacturer’s illustrated parts families; actual configurations must be matched to the machine.

FIGURE T08

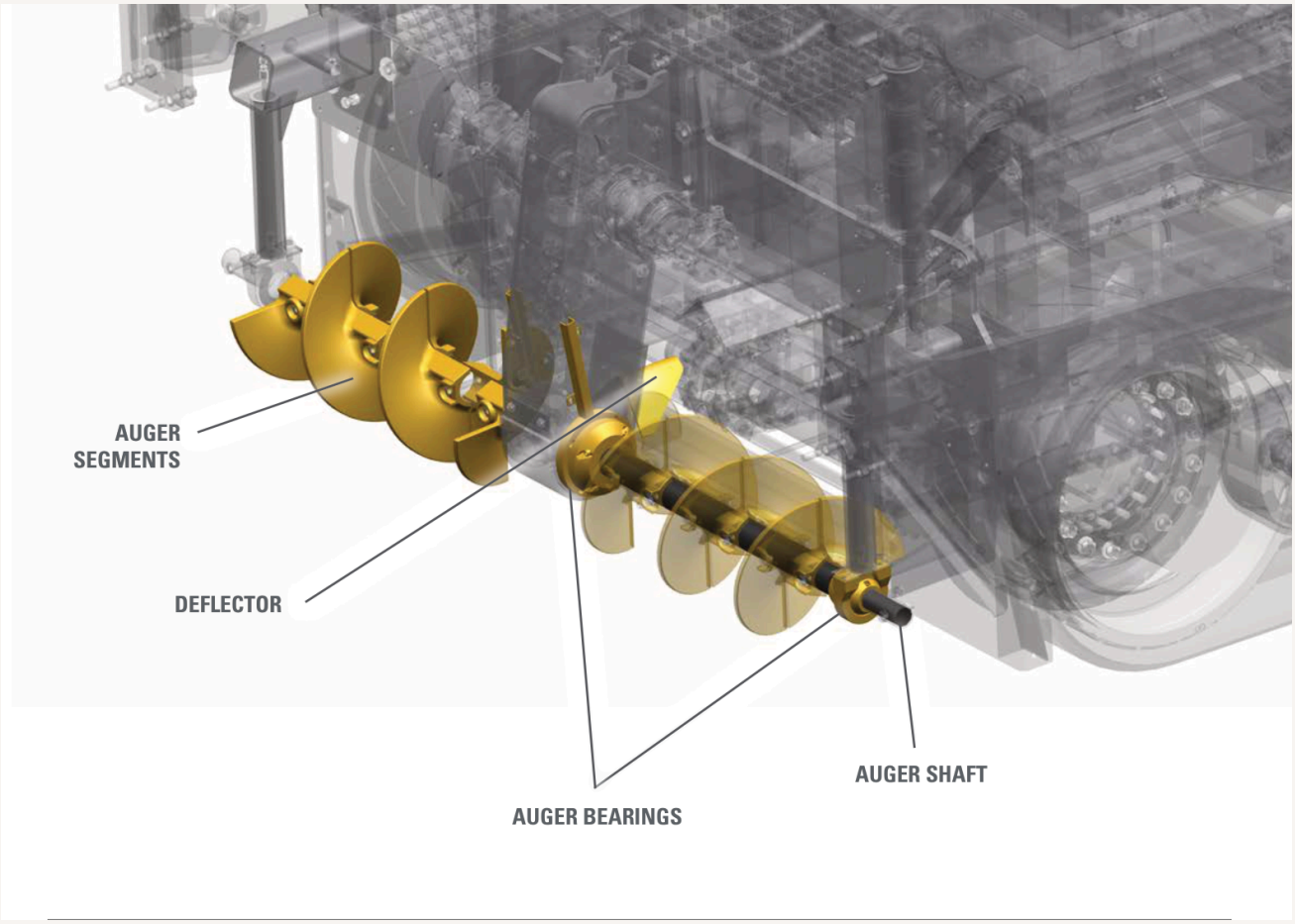
Inside the paver: the feeder system



Note. From Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 11), by Caterpillar, 2024. Copyright Caterpillar. Faithful 200 dpi colour crop; original diagrams, callouts and labels retained. No labels invented or replaced. Complete source-page link accompanies the figure. Scope: Multiple paver/screed models and build numbers; applicability is specified in individual tables. Source and context.

FIGURE T09

Auger parts: identify the assembly



Note. From Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 15), by Caterpillar, 2024. Copyright Caterpillar. Faithful 200 dpi colour crop; original diagrams, callouts and labels retained. No labels invented or replaced. Complete source-page link accompanies the figure. Scope: Multiple paver/screed models and build numbers; applicability is specified in individual tables. Source and context.

TB3-01 · Trace movement and identify parts

On the two CAT illustrations identify feeder bed plate, feeder chain, foot shaft, head shaft, auger segments and auger shaft. Explain which picture concerns rearward feeding and which concerns lateral distribution. Write one sentence distinguishing a wear surface from a moving assembly. This is a picture-identification exercise; inspection or maintenance around these mechanisms requires the actual isolation procedure and qualified instruction. **Skills:** Reading, Writing, Communication.

Read the screed as a responding system

A free-floating screed is pulled by the tractor and responds to interacting forces and material conditions. The handbook drawing names forward motion, screed weight, head of material, angle of attack and material stability. **Head of material** describes the material in front of the screed. **Angle of attack** describes the screed's angle in this system. A change in one condition can affect the others and the resulting mat. A learner should be able to name these relationships before interpreting an adjustment procedure.

FIGURE T10

Read the forces around a free-floating screed

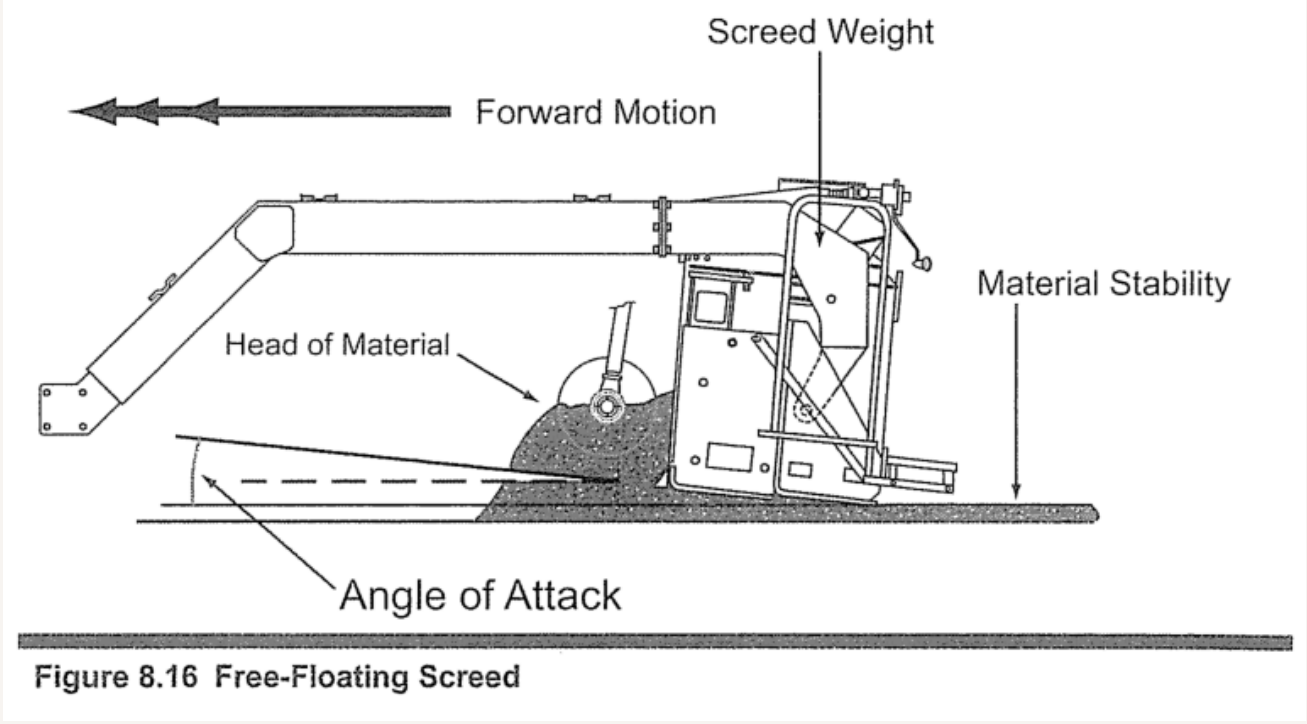
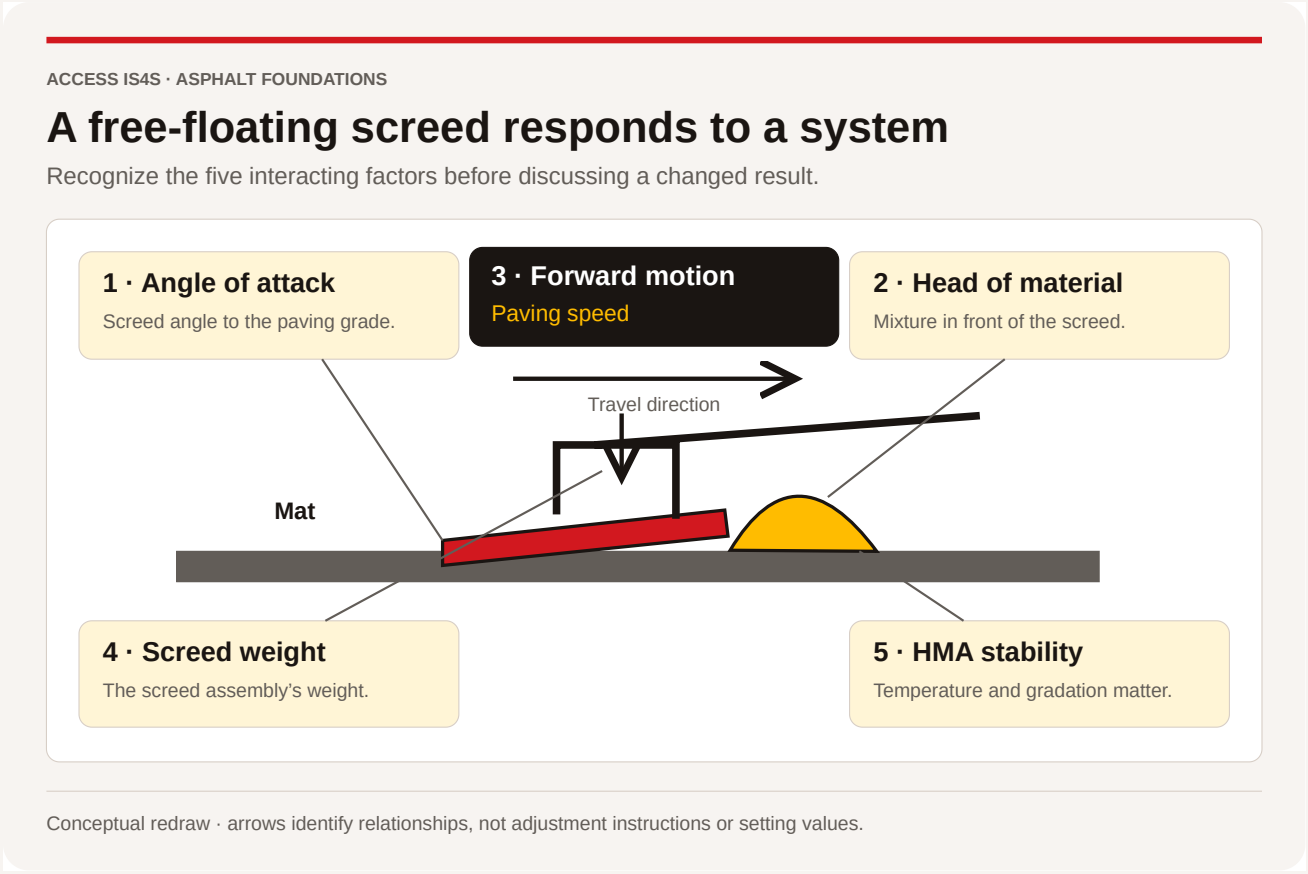


Figure 8.16 Free-Floating Screed

Note. From The asphalt handbook (7th ed., MS-4, Figure 8.16, p. 383; source PDF 417), by Asphalt Institute, 2007. Copyright 2007 by Asphalt Institute. Faithful crop from the supplied scan; no labels altered. Original figure contributor: Volvo Construction Equipment (credit on PDF 804). Source and context.

FIGURE T11

A free-floating screed responds to a system



Note. Adapted from Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4). Figure 8.16, p. 383 / PDF 417. Mirrored and simplified the schematic; grouped the five factors in numbered callouts; omitted adjustment procedures and numeric settings. Original contributor: Volvo Construction Equipment. Copyright 2007 by Asphalt Institute. ACCESS redraw (2026). Source and context.

TB3-02 · Explain a relationship without guessing a setting

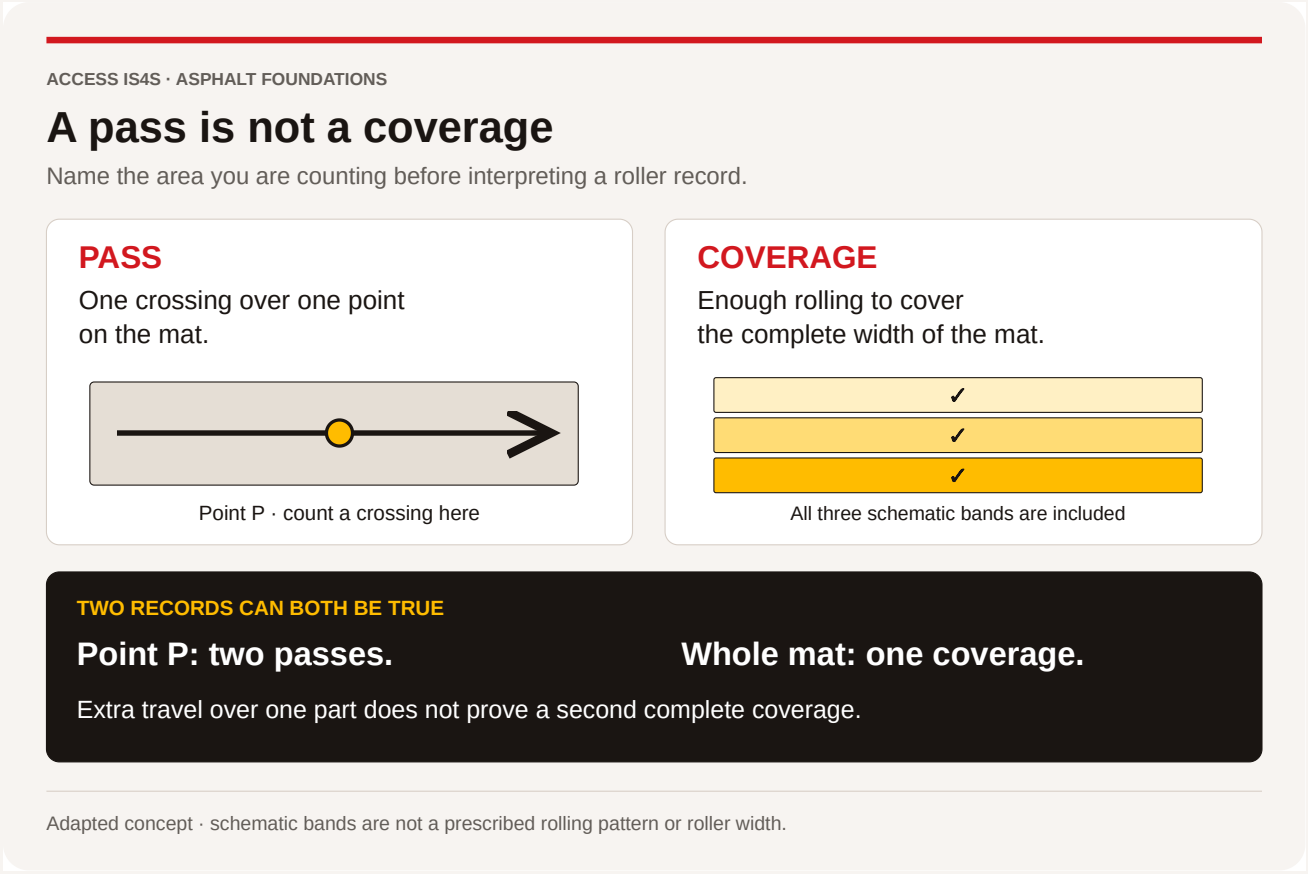
Match five factors in the source picture to the redraw. A fictional learner says, “The screed is heavy, so the layer must always be the same thickness.” Explain why the diagram does not support that conclusion. Identify which machine information and supervised instruction would be needed before changing a control. **Skills:** Reading, Problem Solving, Communication.

Count an observation correctly

A **pass** counts a roller crossing over a point. **Coverage** concerns reaching the full intended width. These are different observations. A record that counts repeated travel over one point cannot establish that every part of the mat received the same work. Neither count replaces the required density evidence.

FIGURE T12

A pass is not a coverage



Note. Adapted from Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4). Figure 9.1, p. 416 / PDF 450. Replaced the original five-pass pattern with separate point and full-width concepts, plus a new record-reading example. No operating pattern is prescribed. Copyright 2007 by Asphalt Institute. ACCESS redraw (2026). Source and context.

TB3-03 · Check the claim in a roller record

Fictional record: point P was crossed twice; the left strip was crossed twice, the centre strip once and the right strip not at all. Can the writer claim complete width coverage? Explain what the record supports and what remains unknown. Design a clearer record with location, time, observed action and a space for the required test reference. **Skills:** Numeracy, Problem Solving, Creativity and Innovation, Writing.

4 · Working in a paving team

SKILLEDTRADESBC · LINE A

A4 · Describe industry expectations of/for workers

Read the trade words, use them in a clear explanation, then keep evidence of your own work.

Outline learning task	Practise here
A4.1 · Explain the importance of Health and Safety	LA4-02
A4.2 · Define types of paving jobs	LA4-01
A4.3 · Explain license requirements for paving equipment operators	LA4-03
A4.4 · Demonstrate professionalism	LA4-04
A4.5 · Explain employer’s responsibility to provide a safe environment	LA4-02
A4.6 · Describe advancement opportunities	LA4-06
A4.7 · Describe job security	LA4-05–06
A4.8 · Explain role of organized Labour	LA4-06
A4.9 · Review working conditions	LA4-05–06

Connect the job title to the work

Outline job term	Broad function to explain	Information another role needs
Raker	Supports hand placement and finishing	Where the agreed handwork is needed
Screed	Forms the placed mat as part of the paver system; the screed role attends to this work	The applicable placement requirements and changes
Asphalt Paver	Places the supplied mixture through the machine system	Delivery status and the approved work arrangement
Compaction Roller	Compacts the placed material under the assigned procedure	Placement status and the applicable compaction requirements
Milling machine	Removes or profiles existing pavement	The authorized removal or profile requirements

Worked language check. “The roller does the road” is hard to check. “The compaction roller compacts the placed material” names the equipment and its function. In LA4-01, make each explanation this specific, then add the information handover. These are broad functions, not machine operating instructions.

Understand the role and the handover

A paving team coordinates people whose tasks depend on one another. The driver delivers material; the paver operator manages the machine's placement function; the screed role attends to the screed's contribution to the placed mat; rakers support hand placement and finishing; roller operators perform the assigned compaction work; milling operators work with equipment that removes or profiles existing material. Job titles and boundaries vary. A job title alone does not establish a person's training, authorization or responsibility on a particular site.

Fictional team brief W1. At the Willow Reach training centre, a classroom team is preparing an explanation of a paving job. Jo coordinates the display. Ren checks the delivery/placement sequence. Ash prepares the screed-and-raker explanation. Kim checks the compaction and milling distinctions. The team has public introductory descriptions, but no site plan, machine manuals or permission to enter a worksite. The display must explain broad roles and show where one role needs information from another.

LA4-01 · Make a role and dependency map

Map haul delivery, paver, screed, raker, roller and milling roles. For each, write one broad function and one information handover. Identify two assumptions W1 does not authorize. Each learner explains a different role before the group combines the map. **Skills:** Reading, Communication, Collaboration.

Treat health and safety as a shared system with defined duties

Health and safety includes planning, suitable equipment and systems, instruction, supervision, reporting and review. It is not satisfied by telling a worker to “be careful.” A new person needs to know the supervisor, relevant hazards, emergency arrangements and how the actual task is done safely. Employers have responsibilities that cannot be replaced by a worker's confidence or a classroom certificate.

B.C.'s orientation provisions require workplace-specific training before a young or new worker begins work. A new worker can be an experienced person who is new to that workplace or affected by changed hazards. Additional orientation and training must be provided when observation shows a need or when the young or new worker requests it. Records are required. The complete workplace program and applicable legislation must guide actual work.

Fictional case W2. A learner arrives for a first workplace visit after completing classroom vocabulary. A host says, “You know the equipment names, so join that crew.” No supervisor contact, task demonstration or emergency information has been provided. Another host offers to stop the assignment and arrange the workplace orientation. The learner wants to participate but does not know the site's communication arrangements.

LA4-02 · Choose the responsible next step

Explain why the second host's approach addresses a real gap. Draft a brief request for orientation and task clarification. Identify information the employer must provide and a question the learner can ask. Do not improvise signals or a route through live equipment. **Skills:** Problem Solving, Writing, Communication.

Check licences and authorization precisely

A driver's licence, admission to a training course, equipment competence and site authorization answer different questions. ICBC identifies Class 5 as the common passenger-vehicle licence. A provider may publish its own admission requirement. Neither statement proves that the holder may operate every paving machine or drive every vehicle combination. Determine the actual vehicle, activity and location, then verify the applicable licence class, restrictions, endorsements, training and employer authorization through the responsible sources.

Document-check case W3, using a real provider statement and fictional decision. IUOE Local 115's public Asphalt Paving Laydown Technician page, retrieved 29 October 2026, lists a valid Class 5 or 7 driver's licence among its admission prerequisites. A fictional learner reads this and says, “Then Class 7 authorizes me to drive any haul truck at work.” The page is a course-admission source; it is not a legal ruling on that vehicle or a workplace assignment.

LA4-03 · Separate four kinds of permission

Create a table for course admission, driver's licence, equipment competence and site assignment. State the question each answers and who or what record can confirm it. Correct the learner's claim without replacing it with another unsupported universal rule. **Skills:** Reading, Problem Solving, Writing.

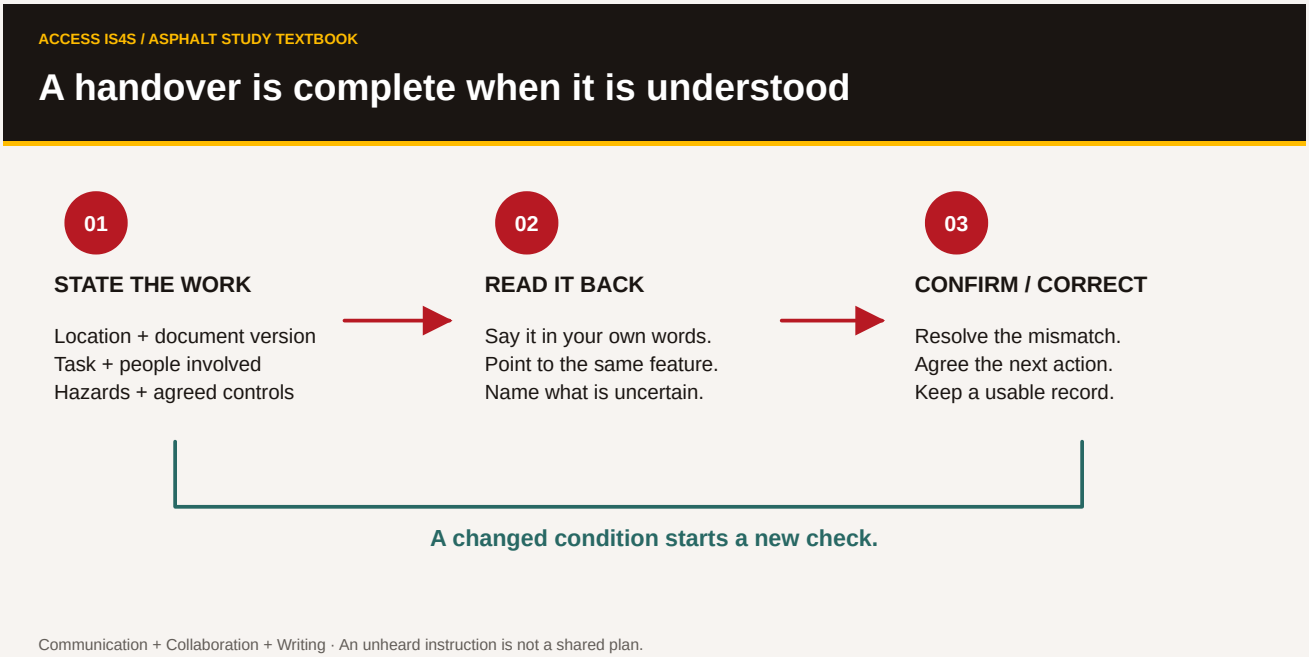
Practise professionalism through observable actions

Professionalism includes preparing for the agreed work, communicating changes, checking information, respecting people and taking care with records and materials. Pride in a project can be shown through a correction made before a mistake is passed on. It does not require pretending to know something or remaining silent when instructions are unclear.

Worked handover. “The map is finished” leaves the receiver guessing. A useful handover is: “The haul, paver and roller functions match our classroom brief. The licensing question is unresolved. The source note is saved as version 2. Please check the role labels; do not describe the draft as a site procedure.” The receiver restates the completed and pending parts before continuing.

FIGURE T13

A handover is complete when it is understood



Note. ACCESS IS4S. (2026). A handover is complete when it is understood [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

LA4-04 · Repair a rushed handover

Jo says, “Everything is approved—just print it,” although W3 is still unresolved. Give a respectful correction that protects the deadline and the accuracy of the work. Agree a revised print note and record each person's contribution. **Skills:** Communication, Collaboration, Adaptability.

Plan development without promising job security

A worker may develop from entry tasks toward equipment roles, specialist functions, supervision or other pathways through training, experience and appropriate assessment. Progress is not automatic or identical for everyone. Ask what evidence, supervision and opportunities the actual employer or provider requires. A forecast of industry demand cannot guarantee a particular person's employment. Season, project timing, skills, location, employer needs and economic conditions can affect work.

Paving work may involve outdoor conditions, dust and fumes, noise, night or weekend schedules and seasonal changes. A job enquiry should ask about the actual schedule, hazards, controls, travel, orientation and employment terms. Do not assume a fictional schedule establishes lawful hours, overtime or pay rules.

Fictional schedule W4. A classroom planning record lists a Monday period from 18:00 to 02:00, including a 30-minute unpaid break for arithmetic practice. Tuesday is cancelled because of a changed project plan. Wednesday has six paid hours. These are invented conditions, not a recommended work arrangement or statement of employment law.

LA4-05 · Check time and revise the plan

Find Monday's elapsed and paid time, then the total paid hours recorded for Monday and Wednesday. Write two questions about a real employer's terms that this classroom record cannot answer. Revise a weekly learning plan so a cancelled day does not become falsely recorded practice. **Skills:** Numeracy, Adaptability, Writing.

Understand organized labour and employment documents

A union can represent workers in collective bargaining with an employer. A collective agreement records negotiated terms for the bargaining relationship; its application depends on the employer, bargaining unit, dates and the actual agreement. B.C.'s Labour Relations Board maintains a searchable directory of filed agreements. An old agreement found online is not automatically the current agreement for a new job. A training provider's fee page is also different from an employment agreement.

LA4-06 · Prepare a useful enquiry and growth plan

Create an enquiry with four questions: applicable employment/collective-agreement information; expected working conditions; training and authorization for a desired role; and how progression is assessed. Add one sentence explaining collective bargaining and one explaining why industry demand or course attendance does not guarantee employment. Add a two-step learning goal with an observable result and a check date. Use fictional names; do not send the enquiry. On a device, save and reopen it with a clear version name. A partner suggests one improvement and you decide whether to use it. **Skills:** Digital, Writing, Creativity and Innovation, Communication.

5 · Read road documents and quantities

SKILLEDTRADESBC · LINE A

A5 · Describe basic road and highway profiles

Read the trade words, use them in a clear explanation, then keep evidence of your own work.

Outline learning task	Practise here
A5.1 · Review categories of projects	LA5-01
A5.2 · Describe types of owners	LA5-01
A5.3 · Describe common types of asphalt work	LA5-03
A5.4 · Review specifications for different types of roads	LA5-01, LA5-05
A5.5 · Describe cross section of road	LA5-02–03

Identify the project before the specification

A project may be described by use—commercial, industrial or residential—or by jurisdiction and delivery, such as a provincial highway project. These categories answer different questions and can overlap. A private industrial site is different from a provincially administered highway, even if both use asphalt. Owners can include provincial, federal, municipal or private organizations. A funder, designer, contractor and owner may be different parties.

Fictional practice project R1 · Cedar Learning Access, drawing P-01 revision B. The fictional Cedar Learning Society owns a short access route on its own practice property. Valley Surface Ltd. is the fictional contractor. The classroom project brief classifies the work as a private facility access project. A fictional community fund contributes money but does not become the owner. There is no real construction instruction or permission attached to this case.

The practice document register contains P-01 revision B, a plan/profile/section sheet; Q-01 revision B, a quantity record; and SP-01 revision B, a teaching specification card. An earlier P-01 revision A is retained for comparison and marked superseded. In a real project, the contract and authorized document system determine applicable requirements; a familiar document title does not establish that it governs every road.

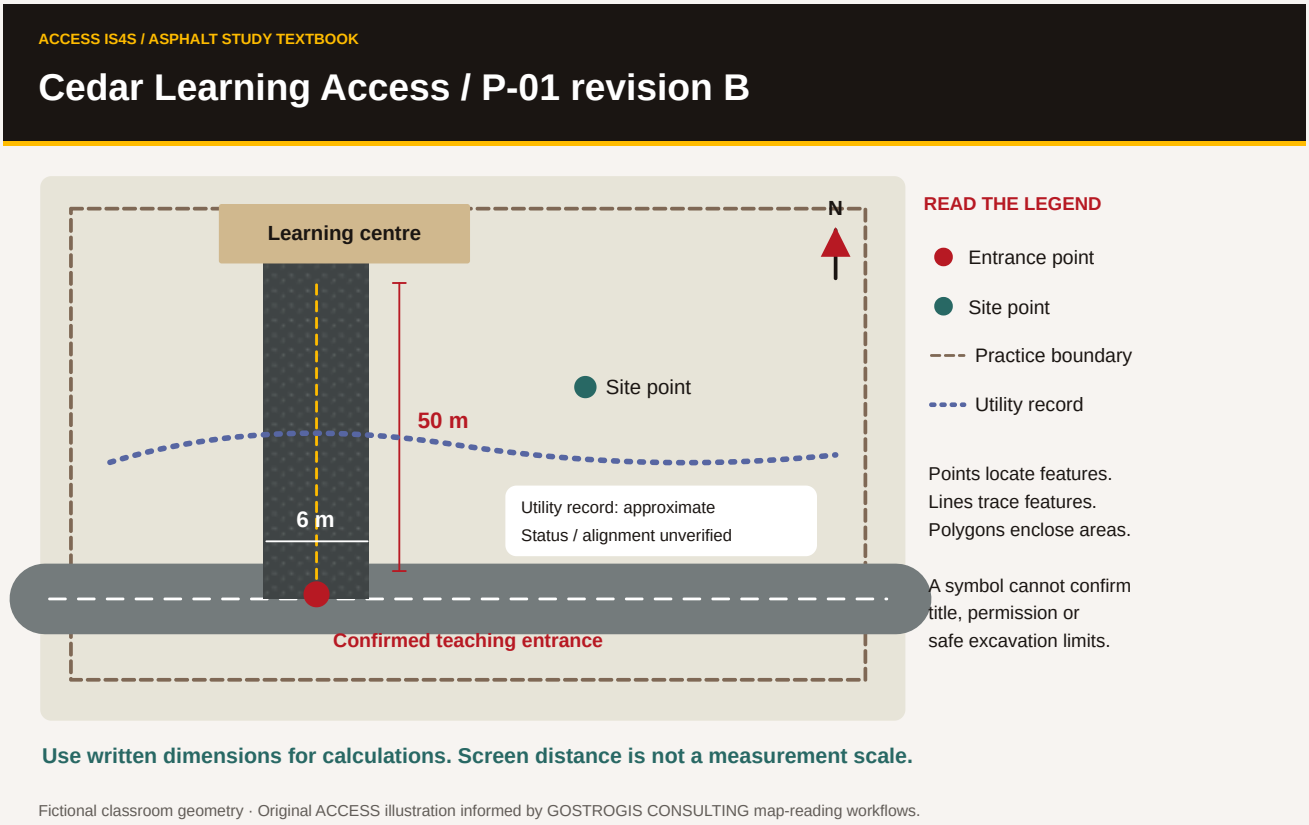
Document families include B.C.'s provincial highway specifications, **Master Municipal Construction Documents (MMCD)**, a local government's project requirements, and building-specific architectural or civil drawings and specifications. MMCD's publisher produces standardized municipal construction documents and updates. A municipal or private project may adopt particular documents through its contract; naming MMCD does not establish that every edition or clause applies. Provincial specifications likewise need the applicable contract version and special provisions. Record the issuer, edition, project adoption and any authorized change before relying on a requirement.

LA5-01 · Sort use, ownership and authority

Identify the project use, owner, contractor and funder in R1. Explain why “private” and “asphalt” do not tell you which public standard or special provision applies. Compare provincial, MMCD, local-government and building-specific document families: what project-adoption evidence would you check before using any of them? Name the record you would need before treating a requirement as authorized. **Skills:** Reading, Problem Solving, Writing.

FIGURE T14

Cedar Learning Access / P-01 revision B



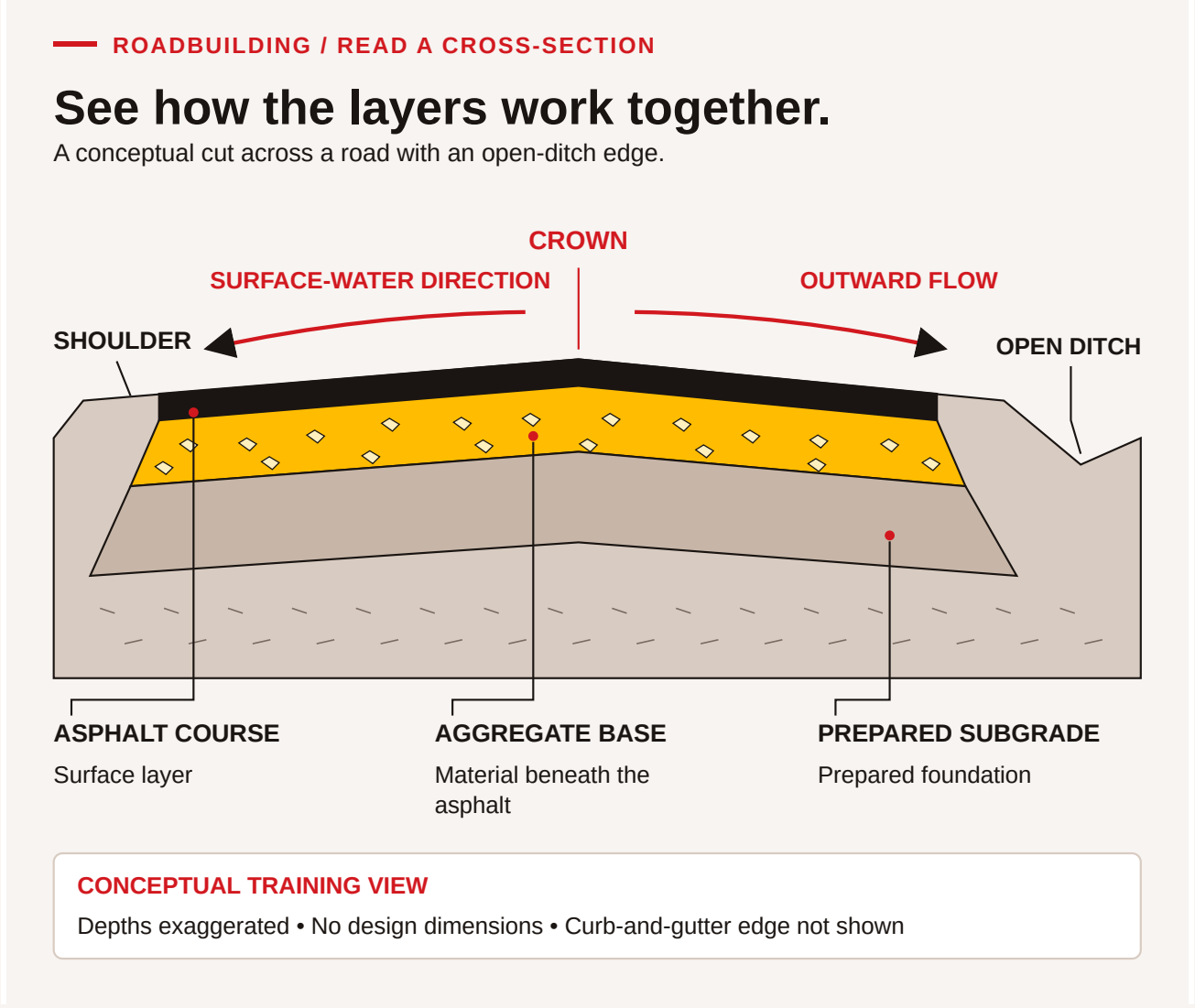
Note. ACCESS IS4S. (2026). Cedar Learning Access / P-01 revision B [Original instructional SVG]. Informed by GOSTROGIS. (2026). Site planner, parcel site planner, map symbology and utility-network documentation (commit 9e409ba). All geometry is fictional and not georeferenced.

Read three views of the same place

A **plan** looks down on a project and shows its horizontal arrangement. A **profile** shows elevations along a chosen line, commonly using chainage or stationing to locate points. A **cross-section** cuts across the route and shows its transverse form and layers. A number must be associated with the correct view, direction and unit. Plan width, profile rise and layer thickness are different quantities.

Figure LA-2

See how the layers work together

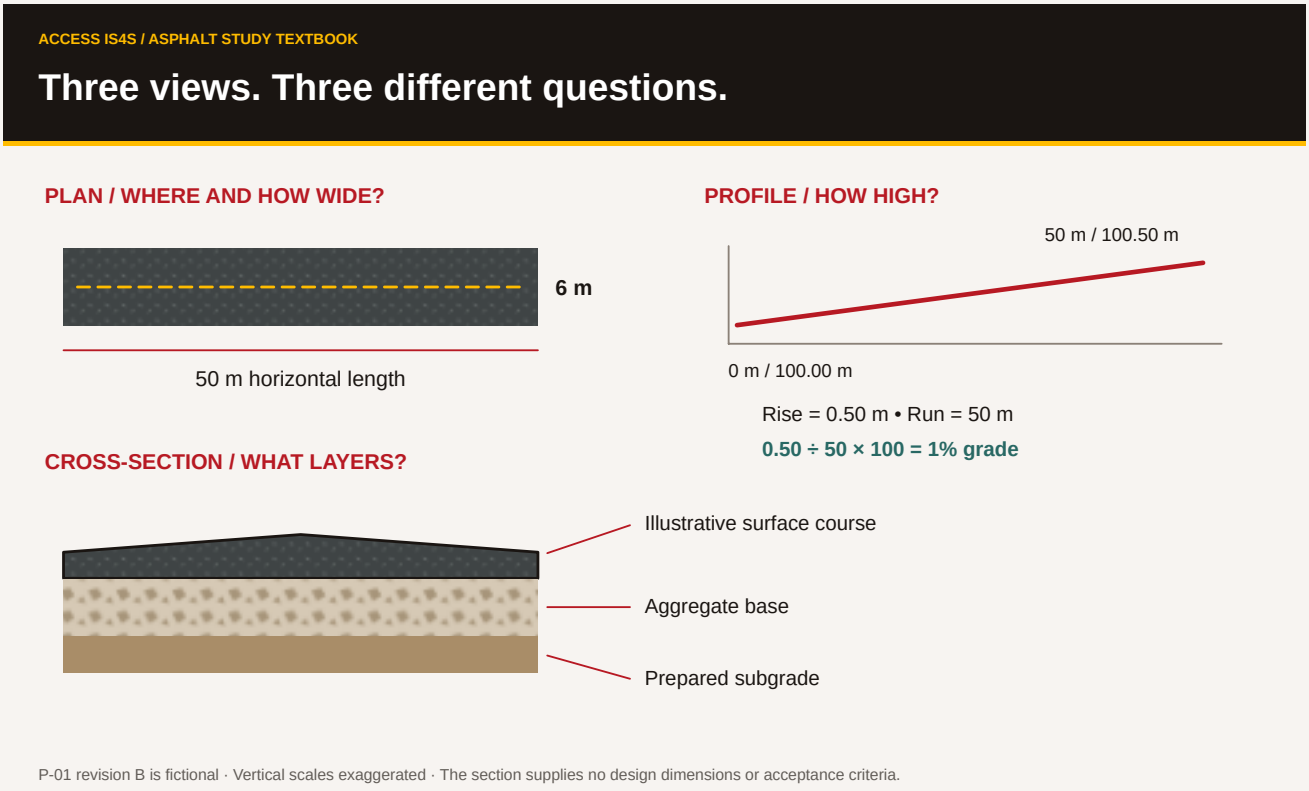


Note. An original ACCESS IS4S instructional illustration. Layer depths are exaggerated. Curb-and-gutter and superelevation alternatives are discussed in the text, not drawn here. No design dimensions or acceptance standards are supplied. By ACCESS IS4S (2026b); see the figure references in the working reference section.

A **crown** is a raised transverse form that can direct water toward the sides. **Superelevation** describes a cross-slope arrangement used on curves; its design depends on the project. **Runoff** is water moving from the surface. An open ditch and shoulder arrangement differs from a contained curb-and-gutter edge. A sidewalk serves a different function from the vehicle surface. A grader is equipment used in earthwork or grading, not a layer in the finished section. Drainage and preparation must be coordinated before paving; a drawing needs more than a surface outline to establish a buildable design.

FIGURE T15

Three views. Three different questions.



Note. ACCESS IS4S. (2026). Three views. Three different questions. [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

Distinguish the supporting layers

A **base** and **subbase** are supporting layers where the design includes them. The **prepared subgrade** is the prepared ground beneath the pavement structure. Figure LA-6 separates one unbound-base arrangement so you can name the layers. Other structures exist, including full-depth asphalt; a subbase is not automatically present on every road. This teaching illustration supplies no dimensions for P-01.

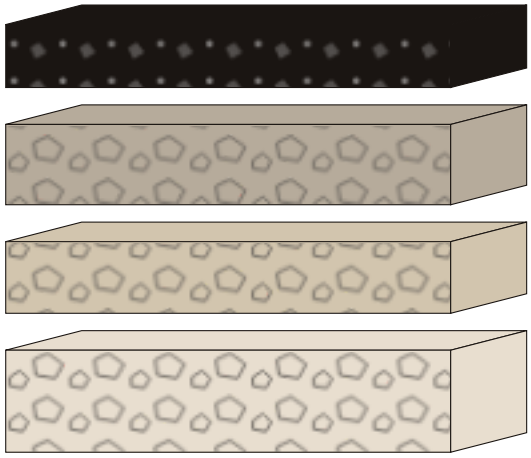
Figure LA-6

Name the layers

ACCESS IS4S · ASPHALT FOUNDATIONS

Name the layers

One example: asphalt pavement on an unbound base, with a subbase.



1. **Asphalt surface**
   The upper asphalt layer.
2. **Granular base**
   Supports the asphalt surface.
3. **Subbase, where specified**
   An additional supporting layer.
4. **Prepared subgrade**
   The prepared ground beneath the structure.

Conceptual section · other pavement structures exist · layer thicknesses are not a design.

Note. Adapted from *The asphalt handbook* (7th ed., MS-4; Figure 1.6, p. 14 / PDF 48), by Asphalt Institute, 2007. Selected the unbound-base example, separated its layers and added functional descriptions. Other structures and design dimensions are not implied. Copyright 2007 by Asphalt Institute; original contributor credits retained.

FIGURE T16

Compare the handbook’s pavement structures

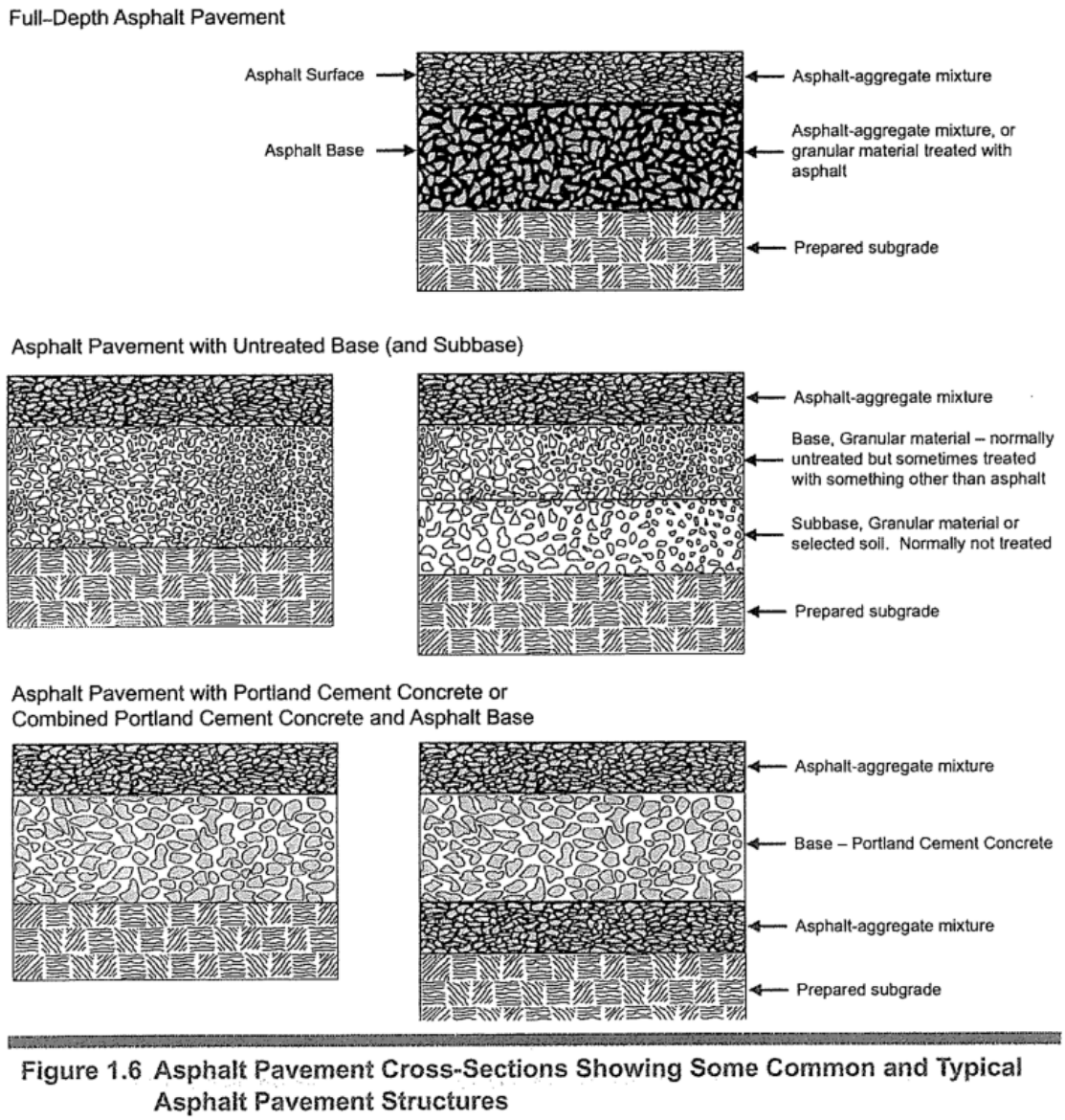


Figure 1.6 Asphalt Pavement Cross-Sections Showing Some Common and Typical Asphalt Pavement Structures

Note. From The asphalt handbook (7th ed., MS-4, Figure 1.6, p. 14; source PDF 48), by Asphalt Institute, 2007. Copyright 2007 by Asphalt Institute. Faithful crop from the supplied scan; no labels altered. Source and context.

P-01 revision B · Complete fictional drawing data. The rectangle extends from chainage 0 m to 50 m. Its constant width is 6 m. The new uniform layer is 40 mm thick for this calculation exercise. No widening, taper, overlap, waste allowance or material loss is included. The exercise asks for geometric volume only. Use the written dimensions; a screen image is not a measuring scale.

Profile point	Chainage along centreline	Illustrative elevation
Start	0 m	100.00 m
Midpoint	25 m	100.25 m
End	50 m	100.50 m

The section names surface course, aggregate base and prepared subgrade. Their presence in a conceptual sketch does not specify material properties, layer thicknesses other than the given calculation layer, drainage design or acceptance criteria. All elevations and dimensions here are invented. The profile arithmetic is not a recommendation for road grade or crossfall.

LA5-02 · Choose the right view

For route length, width, centreline rise and layer thickness, identify the relevant view and unit. Explain why the profile elevation 100.50 m is not the layer thickness. Sketch three boxes labelled plan, profile and cross-section and put the matching information in each. **Skills:** Reading, Numeracy, Creativity and Innovation.

LA5-03 · Describe the road form

Using the visual and paragraph, explain crown, shoulder, open ditch, curb/gutter, sidewalk, runoff and superelevation. Put “grader” in the correct category. Use Figure LA-6 to distinguish granular base, subbase where specified and prepared subgrade; explain why one illustrated arrangement does not establish the layers on every project. Choose one drainage question and one preparation question that the conceptual drawing leaves unanswered. **Skills:** Reading, Communication, Problem Solving.

Point, name and explain

Use the drawing and the reading to connect each word to a function. A learner may point, annotate a copy, or describe the location verbally.

Trade word	Look for or explain	What the word does not tell you
Crown	Raised transverse form near the centre of a crowned road	The specified crossfall or an acceptance result
Shoulder / open ditch	Edge area / open drainage channel	Their required dimensions or capacity
Curb and gutter	A contained edge and associated drainage form	That it is drawn in this open-ditch example
Runoff	Surface water moving away from where it falls	The complete route, quantity or drainage design
Superelevation	A curve-related cross-slope arrangement	That it is the same shape as the crown in this figure
Grader	Equipment used in grading work	A layer in the cross-section
Prepared subgrade	The prepared foundation below the pavement structure	That its preparation has been inspected or accepted

Worked language check. “The arrows are runoff direction; the dark layer is the asphalt course.” The arrows describe movement; the layer describes material. In LA5-03, explain two more features in the same way.

Calculate a quantity you can check

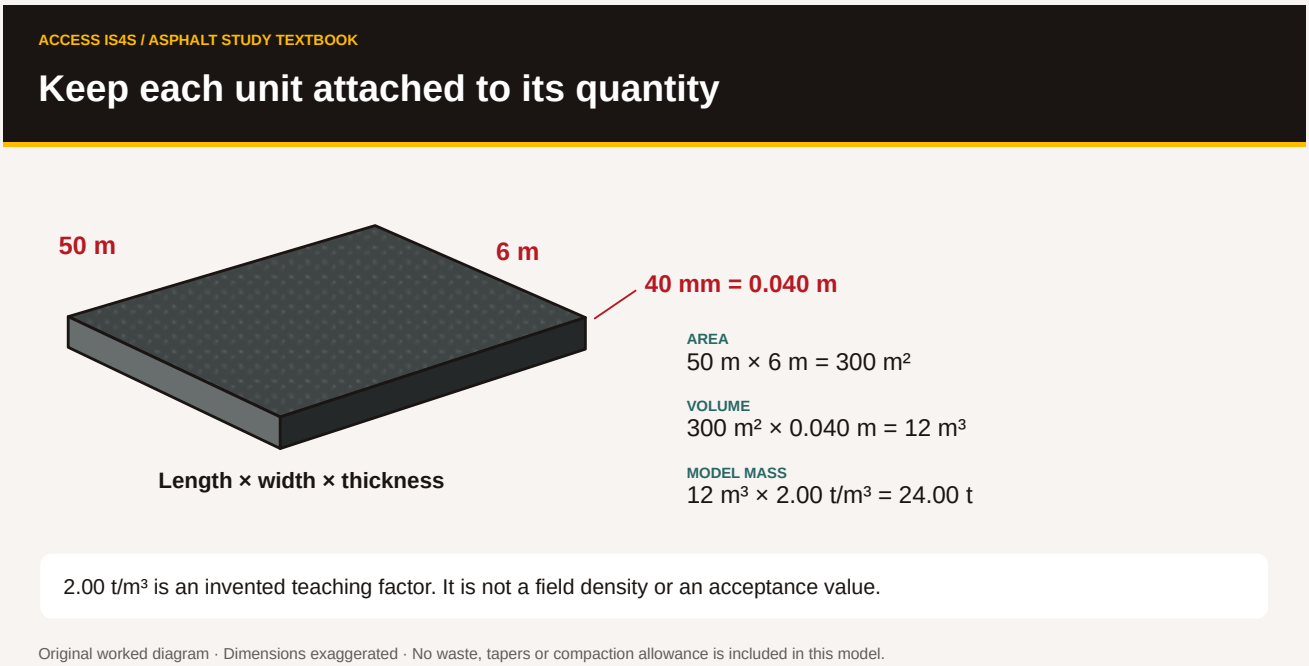
Worked example, P-01 revision B. Convert thickness first: `40 mm ÷ 1,000 = 0.040 m`. Plan area is `50 m × 6 m = 300 m²`. Geometric layer volume is `300 m² × 0.040 m = 12 m³`. Estimate: a 300 m² layer one-tenth of a metre thick would occupy 30 m³; a 0.04 m layer is less than half as thick, so 12 m³ is plausible. Units check as `m × m × m = m³`.

The profile rises `100.50 − 100.00 = 0.50 m` across 50 m of stated horizontal distance. For this drawing-reading exercise, rise/run is `0.50 ÷ 50 × 100 = 1%`. This centreline grade is not the transverse crown or an acceptance limit. A real quantity may need a different geometry and an approved measurement method; our rectangular model has stated simplifications.

Q-01 revision B · Fictional mass model and delivery records. For this exercise only, the instructor supplies a conversion factor of **2.00 tonnes per cubic metre**. It is an invented teaching factor, not an asphalt density specification. The planned model mass is `12 m³ × 2.00 t/m³ = 24.00 t`. Three classroom records list 8.00 t, 8.00 t and 7.60 t. These are fictional record inputs, not a real delivery or placement measurement.

FIGURE T17

Keep each unit attached to its quantity



Note. ACCESS IS4S. (2026). Keep each unit attached to its quantity [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

LA5-04 · Check the quantity and record

Independently reproduce the area, thickness conversion, volume and model mass with units. Sum the three records and calculate their difference from the planned model mass. Explain why that numerical difference alone cannot establish placed thickness, compaction, payment entitlement or rejected work. **Skills:** Numeracy, Reading, Writing.

Use the applicable version and conditions

SP-01 revision B · Teaching specification card, fictional. Use the dimensions printed in the current classroom drawing. Record the revision with every calculation. Keep calculation quantities distinct from acceptance measurements. Refer conflicting instructions to the instructor acting as document coordinator before issuing a final classroom quantity note. This is the complete specification for the exercise; it does not adopt a real road specification.

Change notice R2, fictional. The coordinator issues P-01 revision C, extending the rectangle to 60 m while retaining 6 m width and 40 mm layer thickness. The endpoint illustrative elevation becomes 100.60 m; the same linear 1% profile is used for arithmetic practice. The mass model remains 2.00 t/m³. The three existing classroom records stay unchanged. Revision C replaces B for the new quantity calculation; retain B as the previous version.

LA5-05 · Recalculate after the revision

Find the new area, volume and model mass, and the increase from revision B. Reconcile the unchanged record total with the new model requirement. Explain why changing a filename without recalculating would leave misleading information. Make a versioned paper or digital quantity note with inputs, method, check, result and unresolved limits. **Skills:** Numeracy, Adaptability, Writing; Digital when performed.

LA5-06 · Review and hand over

Work in pairs: one checks geometry and units, the other checks document revision and claims. Swap responsibilities on the revised case. Give a handover that states the current revision, quantity, supplied factor, record total and the decision that remains outside this exercise. Each learner keeps a checked calculation and records one useful correction or confirmed detail. **Skills:** Collaboration, Communication, Problem Solving.

Read the map record behind the symbol

A **point** can identify an entrance or a mapped site location. A **line** can represent a road or a utility record. A **polygon** encloses an area, such as a parcel or project selection. The same coloured dot can be used for very different features; read its legend and attributes. Changing a symbol's colour does not change its position or legal meaning.

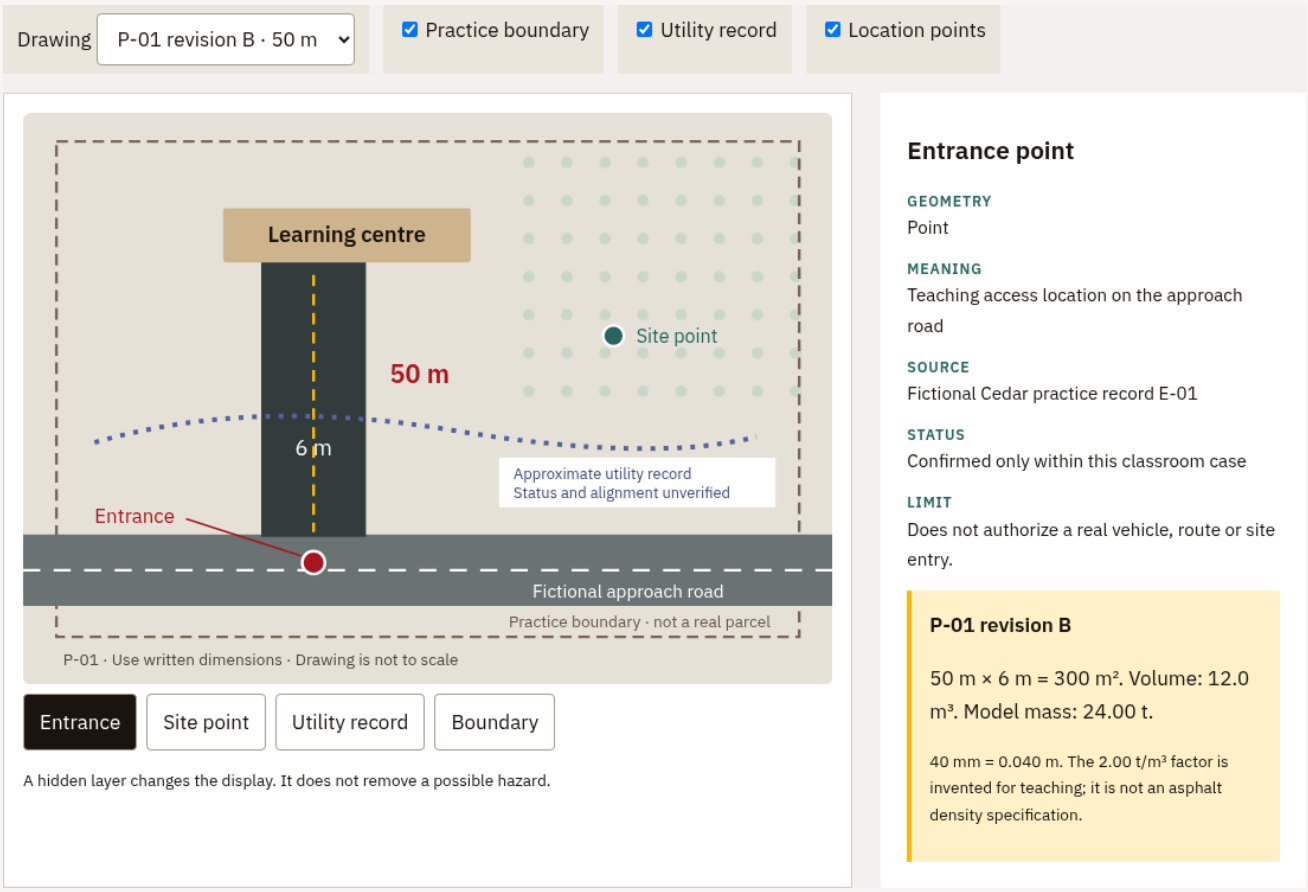
An **entrance** is an access location. A **site point** may locate a facility or represent a record. A **parcel boundary** describes a parcel geometry from a particular source. None of those is interchangeable. A point in the middle of an area does not tell a delivery driver where to enter. A project-selection area filters what is displayed; it does not create or alter a legal boundary.

A useful map-reading record keeps the feature type, source, date, status and uncertainty together. A missing value is not zero. “Unknown utility status” does not mean “no utility.” A source edit date does not prove when a buried line was installed, surveyed or last verified. Separate ownership, operation and maintenance responsibility when the record provides different fields.

Open the Cedar plan desk. It contains fictional features for this lesson. Toggle a layer, select a record, change the drawing revision and export a note. The desk's written dimensions drive the quantity model; no geographic coordinates or real land rights are attached to it.

FIGURE T18

Inspect the record behind the map symbol



Note. ACCESS IS4S. (2026). Cedar Learning Access plan desk [Actual interface screenshot]. Fictional classroom geometry. Map-reading concepts adapted from GOSTROGIS CONSULTING documentation, commit 9e409ba, with owner authorization. Source and context.

Worked map decision. A proposed driver instruction says, “Go to the site dot.” The map shows a site point inside the practice boundary and a separate entrance point on the road. A clearer instruction identifies the confirmed entrance and the current access arrangement. If routing information is used, it must match the vehicle and task: a public car route does not establish truck access, loading restrictions or permission. Road travel distance differs from a straight-line distance between symbols.

TB5-01 · Read, toggle and preserve a record

In the plan desk, select the site point, entrance and utility record in turn. For each, record what it means, its stated status and one thing it cannot establish. Hide and show the utility layer: explain why hiding a feature does not remove the real-world hazard it could represent. Export your own note, reopen the file and check that it retains the selected revision and record. **Skills:** Reading, Digital, Writing, Problem Solving.

TB5-02 · Check the revision in two places

Calculate revision B on paper, then switch the desk to revision C. Compare its written length, area, volume and invented model mass against your LA5-05 result. Explain why the same delivery-record total produces a different difference from the new model requirement. Keep both versions with their labels. **Skills:** Numeracy, Digital, Adaptability.

TB5-03 · Make a usable access brief

Write a 90–120-word brief for the fictional project coordinator. Distinguish the facility, entrance, boundary and approximate utility record. Identify the current revision, the calculation's limited purpose and the information needed before issuing actual work instructions. Ask a partner to identify any claim the map does not support. Improve one sentence and retain both versions. **Skills:** Communication, Collaboration, Writing, Creativity and Innovation.

6 · Investigate a paving trend

SKILLEDTRADESBC · LINE A

A6 · Describe future paving/construction trends

Read the trade words, use them in a clear explanation, then keep evidence of your own work.

Outline learning task	Practise here
A6.1 · Describe green roads	LA6-01, LA6-04
A6.2 · Describe new paving materials	LA6-03–04
A6.3 · Describe new sub-bases	LA6-02
A6.4 · Describe floating roads	LA6-02, LA6-04
A6.5 · Describe added value enhancements	LA6-03, LA6-06

Turn a promising idea into a testable question

A trend is useful to investigate when it responds to a defined need. A product described as *green*, *new* or *innovative* still needs evidence. Ask what problem it addresses, how it works, where it has been studied, what limits apply and who would approve its use. Something new to a local project may have been used elsewhere for decades. Publication date and location matter.

This unit compares material and design ideas without selecting a live road design. You will prepare a recommendation for further investigation using complete evidence cards and a fictional project brief. A careful recommendation can say that one option deserves a closer technical study while leaving design approval to the appropriate professional.

Keep the outline term and explain the mechanism

Outline wording	Meaning to investigate in this unit	First useful question
Green roads	A claim about an environmental purpose or outcome	Which outcome, comparison and time period?
Pave stones / paving stones	Individual paving units in a designed system	Does this particular system manage water through its joints?
Porous asphalt	A surface with interconnected voids, used within an appropriate pavement system	Where will water go, and how will the system be maintained?

Outline wording	Meaning to investigate in this unit	First useful question
Modified asphalt	Asphalt whose binder or mixture has been changed for a defined purpose	Which property changes, and what evidence supports the use?
New sub-bases / pumice	Foundation or fill ideas whose actual design role must be established	Is this material proposed as sub-base, embankment fill or something else?
Floating roads / Styrofoam	Historical outline wording discussed here through engineered lightweight fill and EPS geofoam	What engineered material and geotechnical design are intended?
Street print	A patterned surface treatment or visual enhancement	What are its purpose, performance and maintenance requirements?

A trade term helps you find the right subject. The project documents establish the actual material and use. Keep *sub-base* and *embankment fill* distinct when interpreting the historical cost example below.

Compare surface and reuse approaches

Trend card N1 · Paving stones. Modular paving uses individual units rather than one continuous asphalt layer. Some systems are designed to manage water through joints and underlying layers; the word *paver* alone does not establish permeability. Ask about the whole pavement system, loading, drainage, accessibility and maintenance. A patterned appearance is not environmental evidence.

Trend card N2 · Porous asphalt. FHWA's April 2015 technical brief describes porous asphalt pavements with stone reservoirs as systems intended to provide a surface and manage stormwater. Water moves through interconnected surface voids into the underlying reservoir. This differs from an open-graded friction course on a dense pavement, where water moves through the surface course and out toward the side. The brief identifies potential clogging and the need for specialized maintenance, as well as site/design limitations. A material name does not guarantee infiltration at an unsuitable site.

Trend card N3 · Recycling. Recovered asphalt may enter an approved plant or specialist in-place process. Recovered concrete may be processed as aggregate. Potential reductions in new-material demand need to be examined alongside processing, transport, suitability, maintenance and end-of-life. “Recycled” is a property of the material pathway, not a complete environmental score.

LA6-01 · Improve three claims

Rewrite these statements so they are bounded and useful: “All paving stones absorb rain”; “Porous asphalt never needs drainage planning”; “Any recycled material is automatically the lowest-impact choice.” For each, identify a mechanism, a condition and one question for a qualified reviewer. **Skills:** Reading, Problem Solving, Writing.

Ask what a material changes

Trend card N4 · Epoxy, colour and modified asphalt. Epoxy systems, coloured surfaces and modified binders are families of products and applications. Colour can serve appearance or wayfinding, but it does not by itself establish structural performance, skid properties, durability or accessibility. A modified binder

may be selected for a particular performance requirement. The full product description, intended use, test evidence, compatible construction system and approved specification are needed before adoption. Do not turn a product family into one universal recipe.

Trend card N5 · Lightweight fill and "floating roads." Lightweight fill reduces the load imposed by an embankment compared with a heavier fill arrangement. Expanded polystyrene, or EPS, geofoam is one engineered lightweight-fill material. Its use beneath a road does not mean the road floats on open water. The soil, loading, stability, protection, groundwater and installation conditions require geotechnical design. A brand name or generic word such as *foam* is not a material specification. Pumice is another material encountered in lightweight-fill discussions, with properties and availability that must be checked for the actual source and use.

FHWA's **April 2017 Ground Modification Methods, Volume I** describes lightweight fills and EPS geofoam. It explicitly states that the publication is not a standard, specification or regulation. Its chapter explains that material properties vary and detailed design requires further testing and evaluation. The examples show that lightweight-fill ideas are not all recent inventions.

Evidence table N6 · One historical U.S. cost example. FHWA's 2017 manual reproduces a Milpitas, California project comparison credited to Cheng 2016. The total costs below include material, transport and longevity costs. They **exclude installation and contractor overhead and profit**. They are historical U.S. project figures, not current B.C. quotes. The table's dollar figures are retained in their source context; no currency conversion is supplied.

Material	Reported example total
Traditional soil fill	$563,000
Pumice rock	$633,000
EPS-block geofoam	$1,145,000
Expanded shale/clay	$490,000
Wood chips	$545,000
Tire-derived aggregate	$334,000

Worked example. In this table, pumice's reported total exceeds traditional soil fill by `$633,000 − $563,000 = $70,000`. That result answers a question about this historical comparison. It does not establish that pumice always costs $70,000 more, that the cheapest listed option meets a new site's design needs, or that the table includes every cost.

LA6-02 · Read the cost boundary

Calculate the difference between EPS-block geofoam and pumice in N6. Name three reasons the table cannot be used as a current B.C. quotation. Explain why a lightweight-fill cost table does not by itself establish an approved pavement sub-base material. Explain why price ranking alone cannot establish technical suitability. Keep units and the word *historical* in your conclusion. **Skills:** Numeracy, Reading, Problem Solving.

Separate an enhancement from a requirement

Trend card N7 · Patterned surfaces. Street-print or other patterned treatments can add a visual effect to a pavement. A visual enhancement may serve a project purpose, but performance claims still need evidence. Ask about the actual treatment, friction, wear, maintenance, access needs and compatibility with the underlying system. A decorative pattern is not evidence of structural strengthening or permission to use a cultural design. Any community-specific image or knowledge requires appropriate permission.

LA6-03 · Make a product enquiry

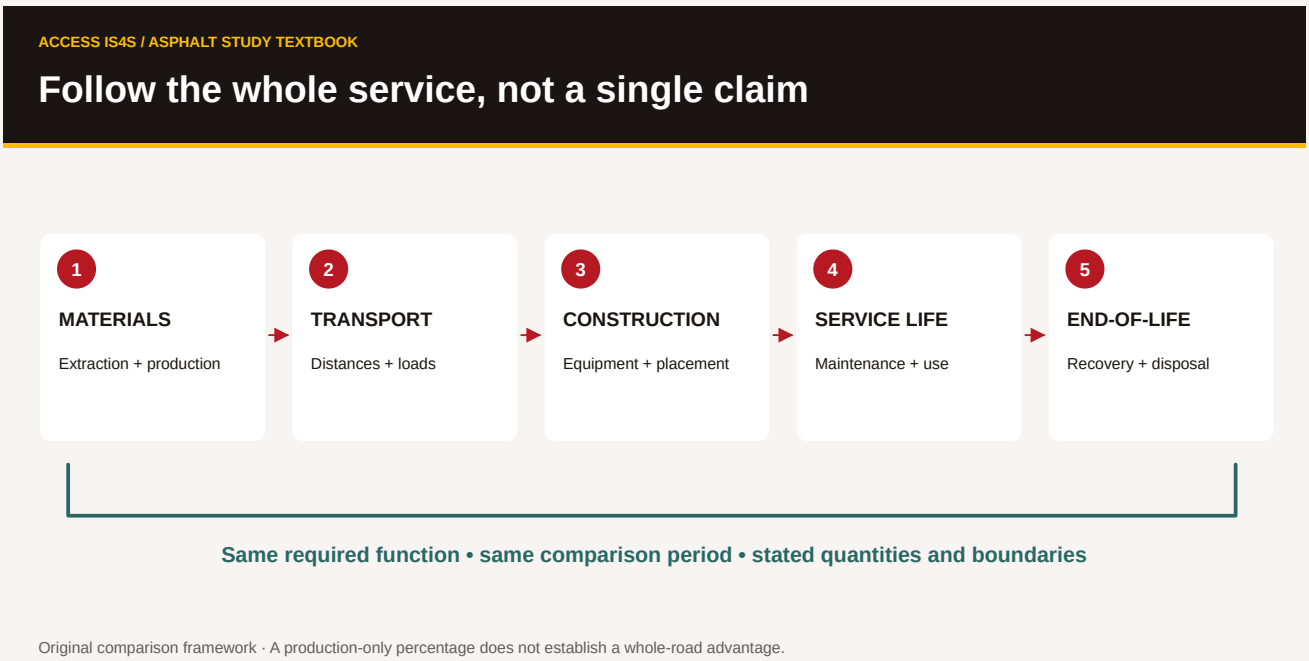
Choose N4 or N7 and draft five questions for a fictional supplier. Include intended use, product identity, evidence, maintenance and approval/permission. Make one question specific enough that a vague marketing answer would be recognizable. Keep the enquiry local; no message is sent. **Skills:** Writing, Communication, Creativity and Innovation.

Apply the evidence to a defined case

Fictional project N8 · A learning-centre courtyard. A facility team is considering a 600 m² courtyard used by pedestrians and occasional authorized service vehicles. The team wants to improve surface-water management and readable wayfinding. No geotechnical investigation, infiltration test, approved loading design, maintenance budget or cultural-design permission has been supplied. The project budget is not yet set. A promoter offers porous asphalt, coloured asphalt or an EPS-supported pavement as three “green” options. These options address different questions and are not interchangeable solutions.

FIGURE T19

Follow the whole service, not a single claim



Note. ACCESS IS4S. (2026). Follow the whole service, not a single claim [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

LA6-04 · Compare options against the actual need

Make a table for the three offers. State the problem each might address, evidence provided in N1–N7, missing project information and the reviewer needed. Select one or two ideas for further investigation, with reasons. Do not approve construction or claim a total environmental winner. Each learner contributes an evidence-based question before the group makes its shortlist. **Skills:** Problem Solving, Collaboration, Reading.

LA6-05 · Adapt to a new constraint

The fictional facility manager says there is no commitment to specialized pavement-cleaning equipment and asks for a maintenance plan before further consideration. Revise your shortlist or investigation plan. Explain why this new fact is relevant to porous asphalt and why it does not automatically approve another option. **Skills:** Adaptability, Communication, Problem Solving.

LA6-06 · Build an evidence brief and test its clarity

Create a one-page brief containing the project question, one proposed investigation, a supported benefit, a limitation, missing information and a next decision. Add source titles/dates from the research record and mark predictions as predictions. Save and reopen the file if using Digital. Ask a partner to identify what you are recommending and what you are leaving undecided; revise one unclear sentence. **Skills:** Writing, Digital, Communication, Creativity and Innovation.

Keep the claims you improved, historical comparison, supplier questions, option table and revised brief. A useful innovation record shows the question, evidence, trial or investigation needed and the limits of the conclusion.

7 · Recognize hazards and prepare for safe work

SKILLEDTRADESBC · LINE B

B1 · Safety orientation

Recognize the hazard, understand the control, confirm the responsibility and explain the next step.

B1 learning task	Classroom evidence
1 · Describe PPE: responsibilities, selection, use and care	TB7-01–02
2 · Describe workplace hazard assessment	TB7-03–04
3 · Describe company safety policies and orientation	TB7-05
4 · Review WHMIS and first-aid requirements	TB7-06–07
5 · Review WorkSafeBC rigging requirements	TB7-08–09
6 · Describe site-specific safety risks	TB7-10–11
7 · Describe public safety and equipment security	TB7-12–13

Begin with the work and the people affected

A **hazard** is a source or situation with the potential to cause harm. **Risk** considers the likelihood and consequences of that harm under the actual conditions. A **control** changes those conditions to prevent or reduce exposure. A control needs an owner, a way to check it and a response if it fails. “Be careful” does not tell a new worker which action is required.

The paving crew works around moving equipment, hot materials, changing surfaces and other people. A task that was controlled yesterday may change with wind, visibility, traffic, equipment condition or a different worker. Before work begins, the crew needs a shared understanding of the task and the current controls. During the work, a changed condition needs attention rather than silent improvisation.

The examples in this chapter teach recognition and decision making. The workplace must provide its own orientation, procedures, training, qualified supervision and emergency arrangements. Read the complete applicable WorkSafeBC requirements when preparing actual work. A classroom exercise does not qualify a person to rig a load, operate machinery, direct traffic or act as the designated first-aid attendant.

7.1 · PPE is selected for a hazard and a person

Personal protective equipment (PPE) is equipment worn to protect a person against specified hazards. Selection starts with an assessment of the exposure and the work. A product that protects against one hazard may not protect against another. A general-purpose glove is different from a glove selected for a particular chemical; eye protection and face protection have different roles; hearing protection must suit the assessed noise exposure and the required program.

Fit and compatibility matter. A respirator must be appropriate for the contaminant and conditions and used within the required respiratory-protection program. A generic “dust mask” description does not establish adequate protection against a gas, oxygen deficiency or a particular dust. A facepiece, eyewear, headgear and other items must work together without making one another ineffective. Training includes the limits of the equipment, not just how to put it on.

Know the provision responsibilities

WorkSafeBC OHS Regulation **8.2** makes the worker responsible for clothing needed for the natural elements, general-purpose work gloves, appropriate footwear including safety footwear, and safety headgear. The employer provides all other PPE required by the regulation at no cost to the worker. Agreements may provide for the employer to supply some or all of the worker-responsibility items. If employer-provided PPE causes allergenic or other adverse health effects, the employer must provide appropriate alternatives or safe measures. Read the actual agreement and the complete regulation before deciding how an item is supplied.

Responsibility to provide an item does not remove the employer's duties concerning suitable protection, instruction and the workplace. Under **8.3**, PPE must provide effective protection, be compatible, avoid creating a hazard and be maintained in good working order and sanitary condition. Where workplace evaluation is required, **8.4** addresses consultation. **8.5** specifies program elements for chemical exposure or oxygen-deficient atmospheres, including selection, use, inspection, cleaning, maintenance, storage and training.

Check	What to establish	Example of a useful record
Hazard and task	What exposure is being controlled?	“Splash task; actual product SDS and procedure identified.”
Selection	Does the equipment protect against this exposure?	Product/model, standard or rating where required, selection confirmation
Fit and compatibility	Can the person use it correctly with other PPE?	Fit/training record as applicable; unresolved interference reported
Inspection	Is it complete, serviceable and within its permitted use?	Defect or condition recorded; unsuitable equipment withdrawn
Use and limits	What can it protect against, and when is it insufficient?	Demonstration and explanation checked by the instructor/supervisor
Care and storage	How is contamination, damage or deterioration prevented?	Manufacturer procedure and assigned care arrangements

TB7-01 · Follow the responsibility without losing the hazard

Sort these fictional supply questions using 8.2: weather jacket, general-purpose work gloves, safety footwear, safety headgear, and chemical-specific protection required by the assessed task. Explain how an employer agreement could affect the first four. Then explain why “I bought it myself” does not establish that an item is suitable. Keep a written answer with the clause and the outstanding selection question. **Skills:** Reading, Writing, Problem Solving.

TB7-02 · Inspect the decision, not just the item

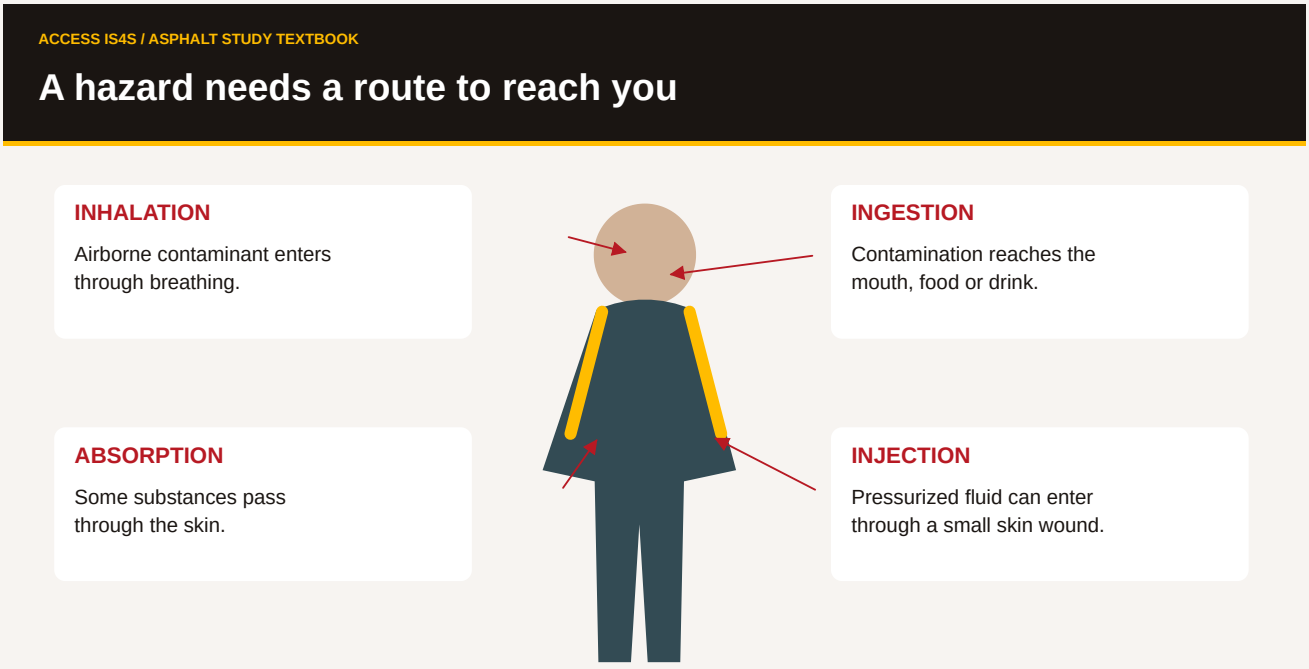
Fictional observations: a learner's eyewear lifts a respirator seal; a glove's chemical suitability is unknown; a headgear component is cracked; a reusable item is visibly contaminated. Make a four-row action record. Identify the unresolved protection issue, how exposure is avoided while it is resolved, who must assess it and what evidence is needed before use. Do not prescribe a substitute product. Explain one case aloud and invite a partner to check it. **Skills:** Communication, Collaboration, Problem Solving.

7.2 · Recognize how exposure happens

A hazard assessment identifies the job steps, people exposed, hazards, existing controls and further controls needed. It must consider routine work and foreseeable changes. Cleaning, clearing a blockage, maintenance, refuelling and a public-access change can introduce hazards that are absent from normal production.

FIGURE T20

A hazard needs a route to reach you



Note. ACCESS IS4S. (2026). A hazard needs a route to reach you [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

Inhalation brings an airborne contaminant into the respiratory system. Carbon monoxide can occur with combustion equipment; hydrogen sulphide can be a concern in particular petroleum-related or confined-space conditions; welding produces fumes specific to the process and materials. Dust-generating work can create respirable crystalline silica exposure. Disturbing suspect asbestos-containing material requires identification and the applicable controls. These examples identify possible hazards to assess, not a claim that every paving site contains every substance. Smell, appearance and a photograph cannot establish a safe airborne concentration.

Ingestion can occur when contamination reaches the mouth through hands, food, drink or other objects. **Absorption** occurs when a substance enters through skin; its significance depends on the substance. **Injection** can occur when high-pressure fluid penetrates skin. A small injection wound can conceal serious injury: report the event and obtain emergency medical care promptly under the emergency procedure. Never use a hand to search for a pressurized leak.

Hazards also act through noise, vibration, heat, cold and physical contact. Noise exposure needs assessment and controls; hearing protection is one part of the required response. Vibration can affect the hands and arms or the whole body, depending on the equipment and exposure. Heat risk depends on workload, weather, clothing and other conditions; a plan may include controls, acclimatization, suitable breaks, hydration and emergency response. Cold and wet conditions can affect health, dexterity and traction. A calendar month alone does not establish a safe exposure.

Biological hazards may include bacteria, other microorganisms or viruses encountered in the actual work environment. Contaminated water, waste or sharps require the relevant assessment and procedure. Workers should report an unexpected condition and keep others from exposure while the responsible person arranges a response. Do not handle an unknown sharp or contaminated material as an improvised classroom demonstration.

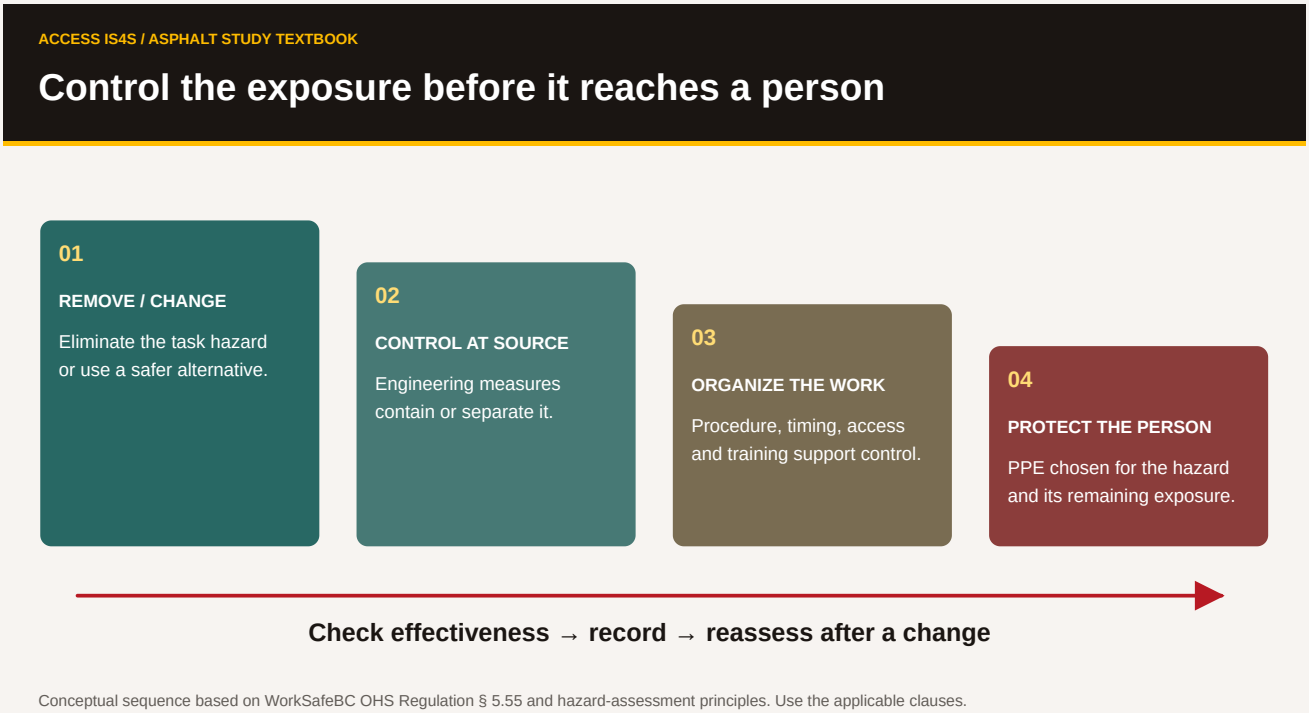
Musculoskeletal injury (MSI) involves muscles, tendons, joints or related tissues. Force, repetition, posture, duration and work organization can contribute. A shovel's load, a rake's reach and a wheelbarrow route affect the demand on a person. A change in tool, load handling, layout or work organization can reduce that demand. An awkward movement may cause a strain or sprain; a crush, impact or fall can cause trauma. Reporting early discomfort or fatigue helps the crew address a developing problem.

Put the control before the exposure

WorkSafeBC **5.55** requires hazardous-substance exposure to be eliminated or controlled below harmful levels and applicable exposure limits using the specified control approaches. A substitute must have known hazards and reduce the risk. PPE as the primary control is permitted only under the conditions in the clause; it is not an automatic first choice. Effective controls often work together.

FIGURE T21

Control the exposure before it reaches a person



Note. ACCESS IS4S. (2026). Control the exposure before it reaches a person [Original instructional SVG]. Based on WorkSafeBC. (2026 retrieved consolidation). Occupational Health and Safety Regulation, § 5.55. Conceptual teaching sequence; use the complete applicable clauses.

Worked example. A fictional crew proposes cutting near an occupied entrance. The observation “dust reaches the doorway” identifies a possible exposure route and affected people. It does not measure the substance or concentration. The next step is to prevent the uncontrolled exposure and have the responsible people assess the material, cutting method, controls and access arrangement. A useful record

identifies the task and location, what was observed, who may be exposed, the interim response, the person responsible and what must be verified before work resumes. Simply handing out unspecified masks leaves critical questions unanswered.

TB7-03 · Build a hazard-and-control record

Use four fictional task cards: (a) combustion equipment proposed beside an enclosed work space; (b) a worker eating with contaminated gloves; (c) repetitive shovelling along an obstructed route; (d) hot weather with a new worker and a changed workload. For each, identify the exposure mechanism, affected people, missing assessment information, a control question and the person responsible for resolving it. State what would trigger reassessment. **Skills:** Reading, Writing, Problem Solving.

TB7-04 · Reassess when the conditions change

Your group has prepared the cutting-task record from the worked example. The facility now needs its entrance for a scheduled delivery, and the wind direction has changed. Revise the record rather than carrying forward its previous conclusion. Create a visual task brief that another group can read in one minute; test whether they can identify the unresolved decisions. Each learner keeps one contribution and one revision. **Skills:** Adaptability, Creativity and Innovation, Collaboration, Communication.

7.3 · Orientation connects policy to a real workplace

A **company safety policy** states commitments and responsibilities. A **procedure** explains how a particular task is controlled. A **manual** may contain equipment or program instructions. **Orientation** connects a person to the workplace: who supervises them, what they are assigned, what hazards apply, what procedures they need and how to get help. Reading a general policy cannot replace a demonstration of the actual task.

WorkSafeBC **3.22–3.25** addresses young and new workers, including workplace-specific orientation and training before work begins, additional training when needed and records. Topics include rights and responsibilities, supervisor contact, hazards, safe work procedures, emergency procedures, first aid and instruction/demonstration of the work. A person experienced elsewhere can still be new to this workplace. Asking for more instruction when a task is unfamiliar is part of responsible work.

Fictional orientation card O-01. Learner: Rowan. Assigned task: document and tool identification under instructor supervision. Supervisor role: classroom instructor. Location: Cedar Learning Access practice drawing. Actual machinery operation: not assigned. First-aid contact, emergency location description and assembly point: to be confirmed for the real training venue. Permitted tools: inspected teaching samples selected by the instructor. Open items: venue emergency arrangements and demonstration boundaries. The card is incomplete until the real venue information is confirmed.

TB7-05 · Close the orientation gaps

Identify what O-01 does establish and what it leaves unresolved. Draft five precise orientation questions, including who supervises, which task is authorized, how help is reached, how a hazard is reported and how additional instruction is obtained. Rehearse a respectful stop-and-clarify conversation. Record the

questions and who must answer them; do not fill venue blanks with invented information. **Skills:** Reading, Writing, Communication.

7.4 · WHMIS and first aid are working systems

WHMIS connects hazard classification, labels, safety data sheets (SDSs), education and workplace training. A **supplier label** identifies a hazardous product and communicates prescribed hazard information. A **workplace label** is required in specified workplace circumstances and has its own information requirements. An **SDS** gives structured information about the product, hazards, handling, exposure controls, first aid and other matters. The label, SDS and actual product identifier must agree.

A pictogram is a starting point for understanding a hazard. It does not give a complete response procedure, exposure limit, glove selection or first-aid instruction. Some classified hazards do not require a pictogram. The environmental symbol may appear on an SDS or label but is not mandatory under WHMIS. Product classification is not determined by choosing whichever picture looks most alarming.

WHMIS SYMBOL STUDY

Read the pictogram with the product information



Flame
Flammability and related hazards



Flame over circle
Oxidizing hazard



Exploding bomb
Explosive / reactive hazards



Gas cylinder
Gases under pressure



Corrosion
Corrosive effects



Skull and crossbones
Severe acute toxicity



Health hazard
Serious health effects



Exclamation mark
Harmful / irritant effects



Biohazardous infectious materials
Infectious biological hazard



Environment
Aquatic hazard; not mandatory under WHMIS

Note. Official pictograms from Canadian Centre for Occupational Health and Safety. (2026). *WHMIS—Pictograms* and official symbol kit. Artwork reproduced unchanged. Short meanings are teaching summaries; classification and conditions require the actual product label/SDS. The environmental pictogram is not mandatory under WHMIS. Complete reference and provenance.

SDS section	What to locate in this chapter
1 · Identification	Product identifier, recommended use and supplier/emergency contact information
2 · Hazard identification	Classification, label elements and other hazards
4 · First-aid measures	Product-specific first-aid information and need for medical attention
6 · Accidental release measures	Spill precautions and response information
7 · Handling and storage	Safe handling, storage and incompatibility information

SDS section	What to locate in this chapter
8 · Exposure controls / personal protection	Applicable control and protection information
10 · Stability and reactivity	Conditions and incompatible materials to avoid

Fictional records exercise W-01. A container is marked “Training Cleaner A.” The available sheet is titled “Training Cleaner B,” revision 2. A colleague says the two are probably similar. No chemical composition, classification or use procedure is supplied in this exercise. The mismatch is the problem to solve; these invented names are not usable products or SDSs.

TB7-06 · Resolve a label-and-SDS mismatch

Explain why W-01 is insufficient for use. Identify the exact information needed and the action to prevent use while the responsible person resolves the mismatch. Use a device to open the local CCOHS label and SDS references from the WHMIS library, find the product-identifier requirement, and save/reopen a short record of what you found. Choose three pictograms from the figure, name their associated hazard meaning and explain why a picture alone cannot select a control. **Skills:** Reading, Digital, Writing, Problem Solving.

First aid depends on the workplace's assessed needs. WorkSafeBC **3.16** requires the employer's written assessment and appropriate equipment, supplies, facilities, attendants and services; related provisions address procedures and communication. A general training certificate does not establish that a particular site has the required coverage. The designated attendant needs the appropriate valid certification for the role, and workers need to know how to summon assistance. The written assessment is reviewed as required and when relevant circumstances change.

TB7-07 · Make the emergency message useful

For the real training venue, identify the confirmed first-aid contact, way to summon help, location description and person responsible for meeting responders. If any item is unknown, record who will confirm it before the activity. In a **spoken simulation only**, practise giving the location, nature of the incident, number of people involved and known immediate hazards. Do not place a real emergency call for practice. Explain why this classroom task does not issue a first-aid qualification. **Skills:** Communication, Writing, Collaboration.

7.5 · Rigging starts with qualified people and known information

Rigging connects and controls a load for lifting or handling using suitable equipment and arrangements. A **sling** connects the load to the lifting system. A **shackle**, hook or other fitting has an identified function, compatibility and working load limit. **Working load limit (WLL)** describes a permitted working load under stated conditions; it is not a universal number detached from configuration, angle or component condition.

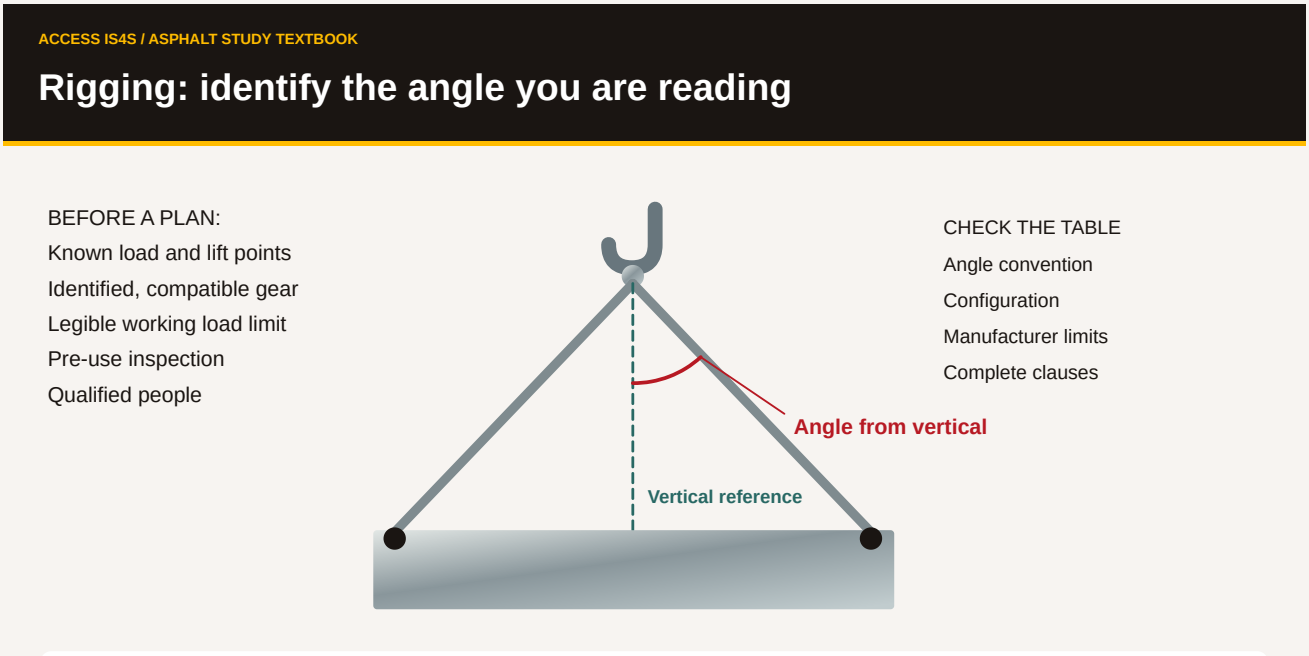
WorkSafeBC **Part 15** is the starting regulatory reading for this task. Under **15.2**, rigging and slinging must be done by or under the direct supervision of qualified workers familiar with the rigging and code of signals. The complete provisions address identification, design factors, WLL, operation, slings, components and

inspection. Manufacturer instructions and the actual lift planning are also needed. A learner must be able to recognize when the information or qualification is missing.

Reading checkpoint	Requirement or question to establish
Qualified person · 15.2	Who is qualified to rig and supervise the proposed work?
WLL and identification · 15.4–15.5	Is the equipment identified and its applicable WLL established?
Signals · 15.20	Which standard signals and communication arrangements apply?
Inspection · 15.31	Slings and attachments must be visually inspected before use on each shift; defective equipment is immediately removed from service.
Sling assembly · 15.33	The WLL of individual components and legs must not be exceeded under the specified rules.
Sling angle · 15.34	Angle is from vertical; required reduction is determined by a qualified person/manufacturer or the table as permitted.

FIGURE T22

Rigging: identify the angle you are reading



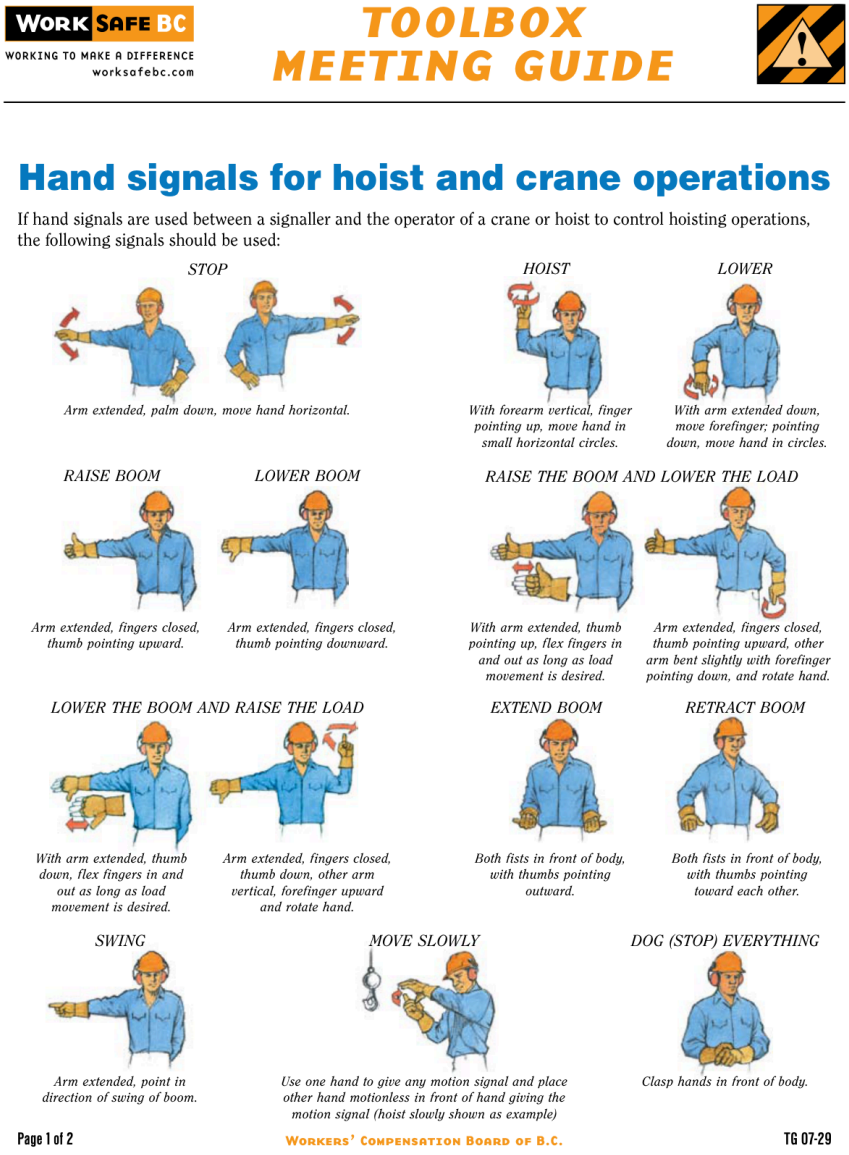
Note. ACCESS IS4S. (2026). Rigging: identify the angle you are reading [Original instructional SVG]. Based on WorkSafeBC. (2026 retrieved consolidation). Occupational Health and Safety Regulation, Part 15, especially §§ 15.2, 15.31 and 15.34. Identifies the angle from vertical; no working load limit or lift arrangement is prescribed.

An angle measured **from vertical** is different from an angle measured from horizontal. A reader must identify the convention before reading a table. The diagram shows the geometry, but supplies no load value, component identification or planned configuration. It cannot be used to select gear. Likewise, a label

on one leg does not establish the allowable load of an entire assembly.

FIGURE T23

Read the established crane signals



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TOOLBOX MEETING GUIDE

Hand signals for hoist and crane operations

If hand signals are used between a signaller and the operator of a crane or hoist to control hoisting operations, the following signals should be used:

STOP
Arm extended, palm down, move hand horizontal.

HOIST
With forearm vertical, finger pointing up, move hand in small horizontal circles.

LOWER
With arm extended down, move forefinger; pointing down, move hand in circles.

RAISE BOOM
Arm extended, fingers closed, thumb pointing upward.

LOWER BOOM
Arm extended, fingers closed, thumb pointing downward.

RAISE THE BOOM AND LOWER THE LOAD
With arm extended, thumb pointing up, flex fingers in and out as long as load movement is desired.
Arm extended, fingers closed, thumb pointing upward, other arm bent slightly with forefinger pointing down, and rotate hand.

LOWER THE BOOM AND RAISE THE LOAD
With arm extended, thumb down, flex fingers in and out as long as load movement is desired.
Arm extended, fingers closed, thumb down, other arm vertical, forefinger upward and rotate hand.

EXTEND BOOM
Both fists in front of body, with thumbs pointing outward.

RETRACT BOOM
Both fists in front of body, with thumbs pointing toward each other.

SWING
Arm extended, point in direction of swing of boom.

MOVE SLOWLY
Use one hand to give any motion signal and place other hand motionless in front of hand giving the motion signal (hoist slowly shown as example)

DOG (STOP) EVERYTHING
Clasp hands in front of body.

Page 1 of 2
WORKERS' COMPENSATION BOARD OF B.C.
TG 07-29

Note. From Hand signals for hoist and crane operations (TG07-29, p. 1), by WorkSafeBC, n.d. Copyright WorkSafeBC. Complete first page reproduced. Compare the regulation's Figure 15-1 and the applicable lift plan. Source and context.

TB7-08 · Read a proposed lift for missing evidence

Fictional request: "Lift the crate with the slings beside it. It should be light enough. The tag on one sling cannot be read. We have not confirmed the load mass, lift points, configuration or qualified rigger." Write a response identifying at least five unresolved items. Explain why guessing the weight or using the clearest-looking sling is inadequate. Locate 15.2, 15.31 and 15.34 in the local regulation and record the requirements in your own words. No lift is performed. **Skills:** Reading, Writing, Problem Solving.

TB7-09 · Identify geometry and communicate clearly

Point to the vertical reference and sling leg in the diagram. Explain which angle Table 15-3 uses. Then, with the instructor, identify two illustrated crane signals on the official sheet and locate their names. Practise naming/recognizing the pictured signals in the classroom only; this exercise does not authorize signalling a crane. Explain how an unclear or conflicting signal would be handled through the actual safe-work procedure. **Skills:** Reading, Communication, Collaboration.

7.6 · Recognize the site and machine hazards

A paving machine has areas that move, heat, pivot or hold stored energy. A stopped engine does not prove all hazardous energy is controlled. Electrical energy, hydraulic pressure, elevated components, springs, gravity, hot surfaces and pressurized systems can remain hazardous. Isolation, lockout, blocking and verification require the applicable procedure and trained people. Do not reach into a feeder, auger or other moving mechanism to clear material.

FIGURE T24

Find the warning before the control



Illustration 5 g01409730

⚠ WARNING

Explosion Hazard! Improper jumper cable connections can cause an explosion resulting in serious injury or death. Batteries may be located in separate compartments. Refer to the Operation and Maintenance Manual for the correct jump starting procedure.

Hot Surface (4)

This safety message is located near the exhaust outlet.



Illustration 6 g03446436

⚠ WARNING

Hot engine components can cause injury from burns. Before performing maintenance on the engine, allow the engine and the components to cool.

Crush Hazard (5)

This safety message is on both sides of the machine near the articulation joint.

Note. Caterpillar. (2022). CB2.5 GC, CB2.7 GC, CC2.7 GC utility compactors: Operation and maintenance manual (M0130633-02, source PDF p. 9). Complete source-page render. Confirm the publication and serial-number applicability before operating. Source and context.

FIGURE T25

Articulation and the steering frame lock

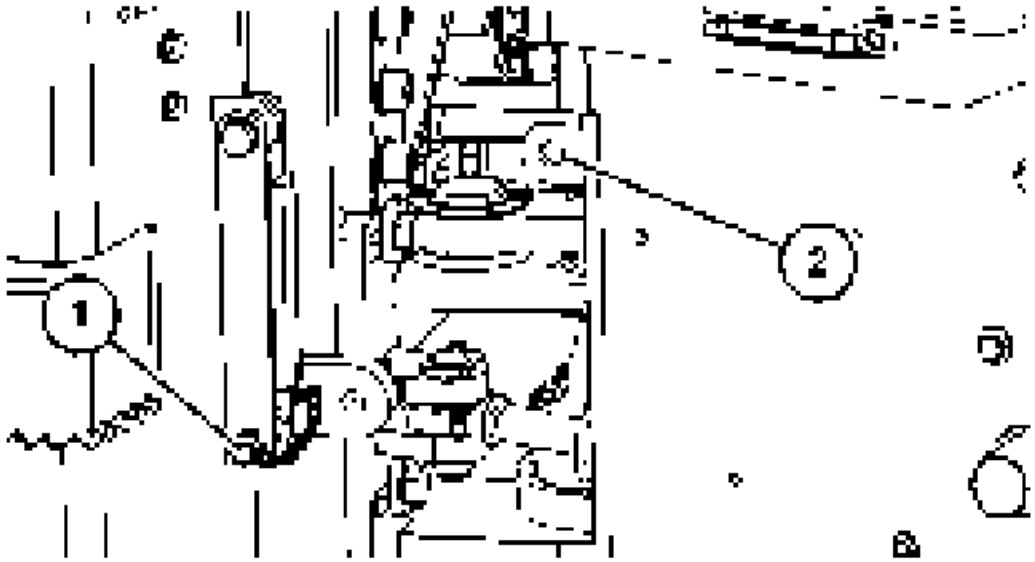


Illustration 34 g06183886

Unlocked position

Note. From CB2.5 GC, CB2.7 GC, CC2.7 GC Utility Compactors — Operation and Maintenance Manual (M0130633-02 (August 2022), source PDF p. 38), by Caterpillar, 2022. Copyright Caterpillar. Faithful 200 dpi colour crop; original diagrams, callouts and labels retained. No labels invented or replaced. Complete source-page link accompanies the figure. Scope: CB2.5 GC: XGC 1-UP, 640 1-UP; CB2.7 GC: BT7 1-UP, 641 1-UP; CC2.7 GC: 2C7 1-UP, 642 1-UP. Read the complete source page for locking/unlocking instructions; this task is identification only. Source and context.

The **articulation joint** allows sections of some machines to pivot. A **steering frame lock** is a model-specific protective device for the circumstances described in its manual. Its existence in a picture does not show that it has been installed or that all energy has been isolated. Batteries, tyres, hydraulic lines and fuel systems require their own procedures. Opening a hot or pressurized system, standing near a moving joint or relying on a raised component without the specified support can cause serious harm.

FIGURE T26

Locate all 19 original roller-control callouts

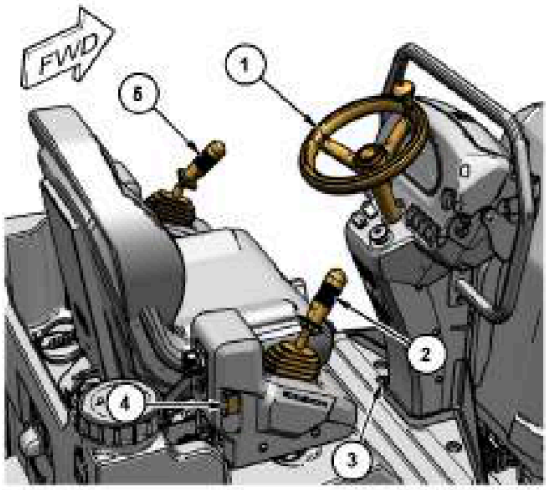


Illustration 39 g06673749

(1) Steering wheel
(2) Right propel control
(3) Traction Control Switch (if equipped)
(4) 12V power receptacle
(5) Left propel control (if equipped)

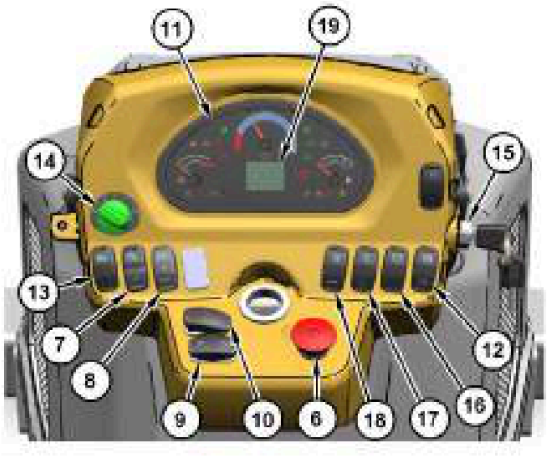


Illustration 40 g06610378

(6) Emergency stop knob
(7) Work lights
(8) Front/Position lights
(9) Turn signal
(10) Hazard lights
(11) Gauges and indicators
(12) Parking brake
(13) Horn
(14) Water spray system
(15) Engine start switch
(16) Vibratory mode / Emulsion switch
(17) Engine speed
(18) Beacon
(19) Digital display

Steering Wheel (1)

Steering wheel (1) controls the directional steering of the machine. The machine will turn in the same direction as steering wheel (1).

Note. From CB2.5 GC, CB2.7 GC, CC2.7 GC Utility Compactors — Operation and Maintenance Manual (M0130633-02 (August 2022), source PDF p. 41), by Caterpillar, 2022. Copyright Caterpillar. Faithful 200 dpi colour crop; original diagrams, callouts and labels retained. No labels invented or replaced. Complete source-page link accompanies the figure. Scope: CB2.5 GC: XGC 1-UP, 640 1-UP; CB2.7 GC: BT7 1-UP, 641 1-UP; CC2.7 GC: 2C7 1-UP, 642 1-UP. Illustration 39 has 1–5; Illustration 40 has 6–19. Optional items retain "if equipped." Source and context.

Look and name. A control panel illustrates why training must match the model. Labels, interlocks and functions vary. Identify the words that you can read in this manual figure, then identify which publication and machine applicability you would verify before touching a real control. The picture is not a starting or shutdown sequence.

Site risk	What must be established before the work
Slips, trips and falls	Access surface, housekeeping, lighting, edges and the applicable fall-protection arrangement

Site risk	What must be established before the work
Moving vehicles, reversing and crush points	Traffic arrangement, visibility, pedestrian separation and agreed communication
Hot mixture, surfaces and burns	Exposure points, separation, suitable protection and emergency response
Overhead utilities	Actual line/voltage information, required limits of approach and the applicable authority's procedure; no single clearance fits every case
Underground utilities	BC 1 Call request and responses, affected owners, verified location and safe excavation procedure; a web-map line is insufficient
Noise and vibration	Assessed exposure, equipment/work controls and required protection/program
Fatigue, illness, drugs or alcohol impairment	Fitness for the task, reporting/support arrangements and reassignment where needed; pressure to continue does not remove the hazard
Inexperience, inattention or complacency	Task-specific instruction, supervision, clear limits and renewed checks after a change

A public utility map can help ask a question. It does not establish a buried asset's exact position, depth, status or clearance for excavation. A record's owner, maintenance responsibility and operational responsibility may be different fields. A record marked proposed or abandoned must not be silently treated as a verified in-service line. Chapter 5's map desk teaches these distinctions with fictional data.

TB7-10 · Use the manual to ask the right question

In the roller safety page, find one warning and identify the hazard and stated response. In the steering-frame illustration, identify the lock and the joint it relates to. State the publication/model context, what the image establishes and what must be verified on the actual machine. Explain why a key switched off or a visible lock is insufficient evidence that every energy source is controlled. **Skills:** Reading, Problem Solving, Writing.

TB7-11 · Stop an unsupported site assumption

Fictional messages: “The utility line on the map is exact”; “The overhead wire looks high enough”; “The engine is off, so the raised part is safe”; “We've done this for years, so no new check is needed.” Rewrite each as a precise verification question and an appropriate interim response that avoids exposure. For one case, use the plan desk and save a note identifying the source status and the missing information. **Skills:** Adaptability, Digital, Writing, Problem Solving.

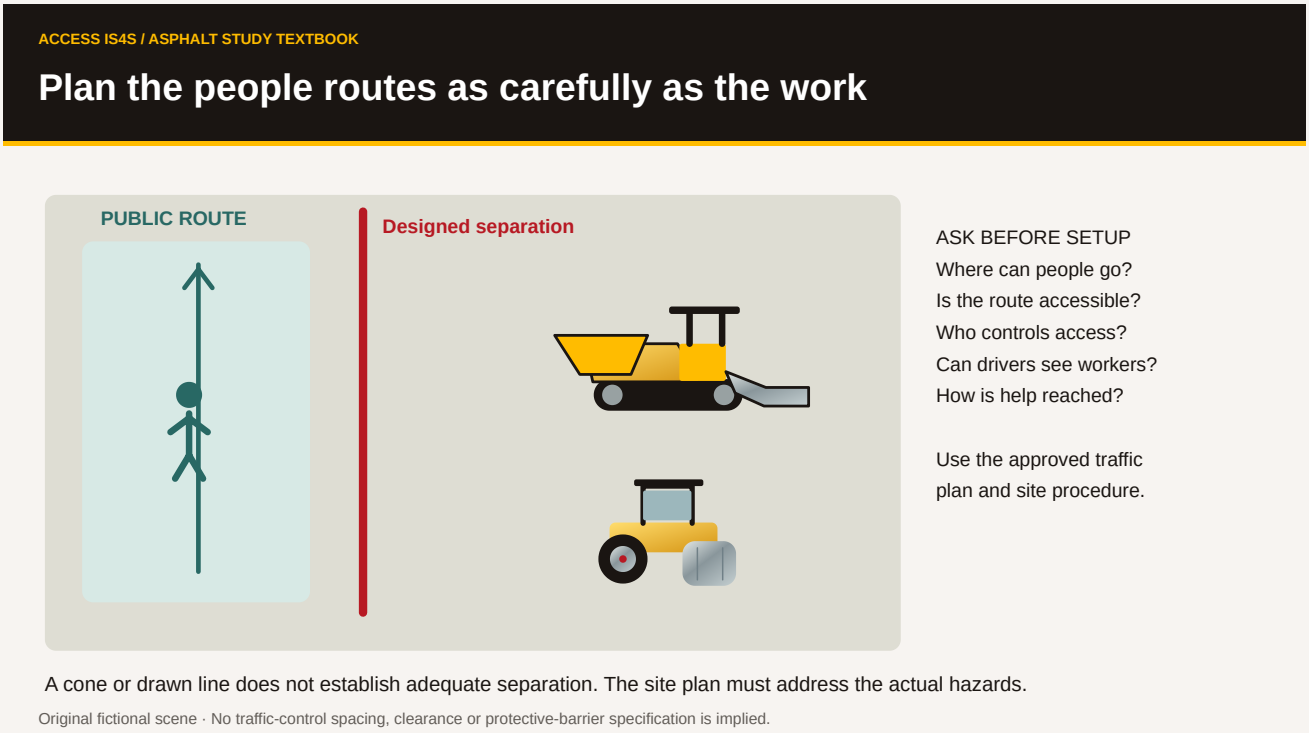
7.7 · Protect the public and secure the equipment

A site can affect people who never joined the crew briefing: visitors, delivery drivers, people using mobility aids, nearby workers and people passing the entrance. The work plan must address how they move, how they are kept from hazards and how information reaches them. Traffic control follows the applicable road

authority's plan and WorkSafeBC Part 18 requirements. A hand-drawn arrow or cone in a teaching picture does not establish a compliant layout.

FIGURE T27

Plan the people routes as carefully as the work



Note. ACCESS IS4S. (2026). Plan the people routes as carefully as the work [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

At the end of a work period, the crew still has responsibilities. Equipment must be parked and secured under the actual manufacturer and site procedures. Brakes, wheel chocks where required, isolation and the treatment of raised components depend on the equipment and conditions. Remove access to unauthorized operation and verify the planned public/site boundaries. A generic instruction to “turn it off” leaves stored energy, movement and access questions unresolved.

Worked example. A crew has secured a machine under its procedure, but the public route now passes beside a hot component. Completing one equipment-security task has not resolved the public-exposure hazard. The responsible people must reassess the access arrangement, control the exposure and communicate the changed route. The handover records both the machine status and the remaining site condition.

TB7-12 · Review the public route

Use the conceptual diagram. Identify who may need access, how a pedestrian/vehicle conflict could arise and what the drawing cannot establish. Create five questions for the responsible site planner, including accessibility, visibility, control of entry, emergency access and changing conditions. Improve the clarity of the public-route notice without inventing a traffic-control layout or barrier specification. **Skills:** Communication, Creativity and Innovation, Problem Solving.

TB7-13 · Integrated classroom safety brief

Prepare a two-page brief for the fictional Cedar Learning Access site. Include: task and limits; PPE decision record; four hazard/control questions; orientation gaps; label/SDS match check; first-aid contact verification; rigging prerequisites; utility information limits; and public/equipment-security handover. Reuse your earlier work but resolve contradictions. List the responsible role and needed evidence beside each open item. Attach your checked Chapter 5 quantity/revision sheet and the angle-convention diagram from TB7-09. Add a visual that helps a classmate find a critical question quickly.

Exchange briefs. Your partner tests whether they can find the unresolved issues without your explanation. Revise, save and reopen your final brief. Each learner presents one decision and one change they made after feedback. **Skills:** all nine Skills for Success. The evidence is your accurate reasoning and clear record, including the points that still require qualified or site-specific confirmation.

Working reference and evidence record

Words used in this book

Term	Meaning in these lessons
Aggregate	Mineral material such as sand, gravel or crushed stone
Asphalt binder / asphalt cement	Binding material in an asphalt mixture; distinct from Portland cement
Batch plant	Production arrangement that makes identifiable batches
Drum-mix plant	Production arrangement using a continuous material/mixing process; layouts vary
RAP	Reclaimed asphalt pavement material considered for an approved reuse pathway
Hot-in-place recycling	Specialist process treating existing asphalt pavement on the road
Screed	Paver component that contributes to forming the placed mat
Compaction	Densification process performed according to the applicable approved procedure
Milling	Removal or profiling of existing material using suitable equipment
Chainage	Position measured along a defined project line
Plan / profile / cross-section	Top view / elevation along a line / cut across a route
Crown	Raised transverse road form that can direct surface water toward sides
Superelevation	Cross-slope arrangement associated with a curve; design is project-specific
Runoff	Water moving from a surface
Specification	Documented requirements; applicability depends on its legal/contractual context
Special provision	Project/contract-specific provision that may modify or add requirements
Revision	Identified version of a document
Functional comparison	Comparison based on an equivalent required service, period and conditions
Life-cycle boundary	Stages included and excluded in an environmental assessment
Percentage point	Difference between two percentages expressed on the same basis
EPS in the 2006 specifications	End Product Specification, explicitly defined in clause 502.03.01; identify the edition and context before applying the term
EPS geofoam	Expanded-polystyrene material for an engineered geotechnical application; an abbreviation's meaning still depends on context

Read the word, find the evidence, explain the limit

To practise a term, write the trade word, point to its feature or source statement, and explain the function in one plain sentence. Then say what the picture or statement cannot establish. Keep technical words intact while making the explanation easier to follow. For example: “The **asphalt course** is the surface layer shown here; this picture does not specify its required thickness.”

Record evidence without overstating it

For each selected activity, record: activity ID; date; task and document version; what you did; support used; result; check; feedback; revision; and next practice. Mark the observed skill. A group product needs an individual contribution record. An accurate calculation needs units and assumptions. A digital claim needs actual tool use. A future application is a plan until it has been observed.

Activity	Evidence kept	Support and feedback	Next practice
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Sources and continuing questions

The research register provides public titles, issuers, dates, URLs, retained short excerpts and retrieval hashes. It distinguishes dated factual evidence from invented classroom inputs and records questions that need current local or technical confirmation. Use the register to check a claim, then keep the question and the evidence together. A link or a confident statement alone is not the complete answer.

Supplied technical document

British Columbia Ministry of Transportation, Construction and Maintenance Branch. *2006 Design Build Standard Specifications for Highway Construction*, adopted 20 February 2006. Section 502.03.01, PDF 297 / printed 1 of 22, defines End Product Specification (EPS). The supplied historical document is distinct from the current public specifications page below; its presence does not establish current contract applicability.

Public reference list

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The curated evidence register gives the teaching use and limits; exact selected excerpts and hashes support checking. The road-industry table extract preserves the nine source rows.

Trade-outline and figure references (APA 7)

SkilledTradesBC. (2016). *Asphalt paving/laydown technician program outline* (August 2016). Program outline, Line A, printed pp. 20–25 / PDF pp. 22–27.

ACCESS IS4S. (2026a). *Follow the material, step by step* [SVG instructional illustration]. Local illustration.

ACCESS IS4S. (2026b). *See how the layers work together* [SVG instructional illustration]. Local illustration.

Figures LA-1 and LA-2 are original ACCESS IS4S teaching illustrations. Figure LA-3 retains the manufacturer's component callouts and is accompanied by a selected text legend. The trade terms are connected to the supplied outline; source-specific technical claims retain their evidence cards and reference entries.

LeeBoy. (2026). *LeeBoy model 8680 commercial paver: Safety and operations manual* (Manual No. 1033428-SOP). https://www.leeboy.com/wp-content/uploads/2024/03/1033428-SOP-MANUAL-8680-REV-A_WEB.pdf

Figure LA-3 uses Figure 3-1, printed p. 3-7 / PDF p. 43. The manufacturer names shown beside the crop are a selected subset of the complete legend. The image provenance manifest records the original PDF hash, page and crop boundary.

Asphalt Institute. (2007). *The asphalt handbook* (7th ed., MS-4). Complete local reader.

Figures LA-4–LA-6 are ACCESS redraws of verified MS-4 figures. LA-4 uses Figures 7.2 and 7.4 (pp. 310, 312 / PDF 344, 346; original contributors Terex Roadbuilding and National Highway Institute). LA-5 uses Figure 8.11 and component descriptions (pp. 377–378 / PDF 411–412; original figure contributor Volvo Construction Equipment). LA-6 uses one unbound-base arrangement from Figure 1.6 (p. 14 / PDF 48). Each caption identifies the simplification. The figure provenance manifest retains the source identity, asset hash and contributor-credit page. The chapter map connects the verified book contents to the occupational outline.

Safety and map-reading references

WorkSafeBC. (2026, retrieved October 30). *Occupational Health and Safety Regulation*, Parts 3, 5, 7, 8, 10, 14, 15, 16, 18, 19 and 20. Retained full-text publisher articles; retrieval date identifies the consolidation consulted, not the date each rule was enacted.

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Canadian Centre for Occupational Health and Safety. (2026). *WHMIS: Pictograms; General; Labels; Safety data sheets*. Retained current fact sheets and official symbol kit. Revision dates and downloaded-file hashes are recorded in the local manifest. Symbols are reproduced unchanged; the environmental symbol is not mandatory under WHMIS.

BC 1 Call. (n.d.). *Six steps to safe digging*. Retained publication. Follow actual utility-owner responses and the applicable excavation procedure.

Canadian Centre for Occupational Health and Safety. (2026 retrieved editions). *Hand tool ergonomics; Shovelling; Powered hand tools—Pneumatic tools; Fatigue*. Complete source catalogue. The catalogue records the individual publication dates where supplied.

Seymour Midwest. (2026 publisher download). *Partner resource guide*. Retained complete catalogue. Product examples illustrate names and configurations; match actual product identifiers before use.

Caterpillar. (2016). *Guide to asphalt paving* (QEBQ1888). Complete retained guide.

Caterpillar. (2022). *CB2.5 GC, CB2.7 GC, CC2.7 GC utility compactors: Operation and maintenance manual* (M0130633-02). Manual catalogue. The figure register and original catalogue retain publication, model and serial-range applicability for every selected machine illustration.

GOSTROGIS. (2026). *Site planner; Parcel site planner; Map symbology; Utility-network coverage; Project areas* [Repository documentation, commit 9e409ba9d5bc28e7da3d400671ec6e7707615613]. Pinned reuse record. Concepts adapted with the repository owner's express authorization. Five authored documentation snapshots are retained locally; no real cadastral, utility, client or student dataset was imported. The Cedar case is fictional.

Additional trade words for safe work

Term	Meaning in this volume
Hazard / risk / control	Potential source of harm / likelihood and consequences in context / measure that prevents or reduces exposure
PPE	Personal protective equipment selected for the assessed hazard, person and task
SDS	Safety data sheet for the identified product
Inhalation / ingestion / absorption / injection	Entry by breathing / swallowing / through skin / penetration such as high-pressure fluid
MSI	Musculoskeletal injury involving muscles, tendons, joints or related tissues
Orientation	Workplace-specific introduction, instruction and confirmation of responsibilities and arrangements
Rigging / sling / WLL	Load-handling connections and arrangements / lifting connector / working load limit under stated conditions
Articulation	Relative pivoting between sections of a machine designed for that movement
Hazardous energy	Energy that can cause harm, including motion, pressure, gravity, electricity and heat
Point / line / polygon	Map geometry locating a feature / tracing a feature / enclosing an area
Status / uncertainty	What a record says about a feature / what remains unknown or unverified

The figure register identifies each picture, its source page or original drawing, transformations and image hash. Captions travel with the figures in the study textbook. The teaching and worked-answer guide includes the coverage map and evidence criteria.